Autonomous Racing - Experiences from Roborace, the Indy Autonomous Challenge and Abu Dhabi Racing League

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Dominik Kulmer

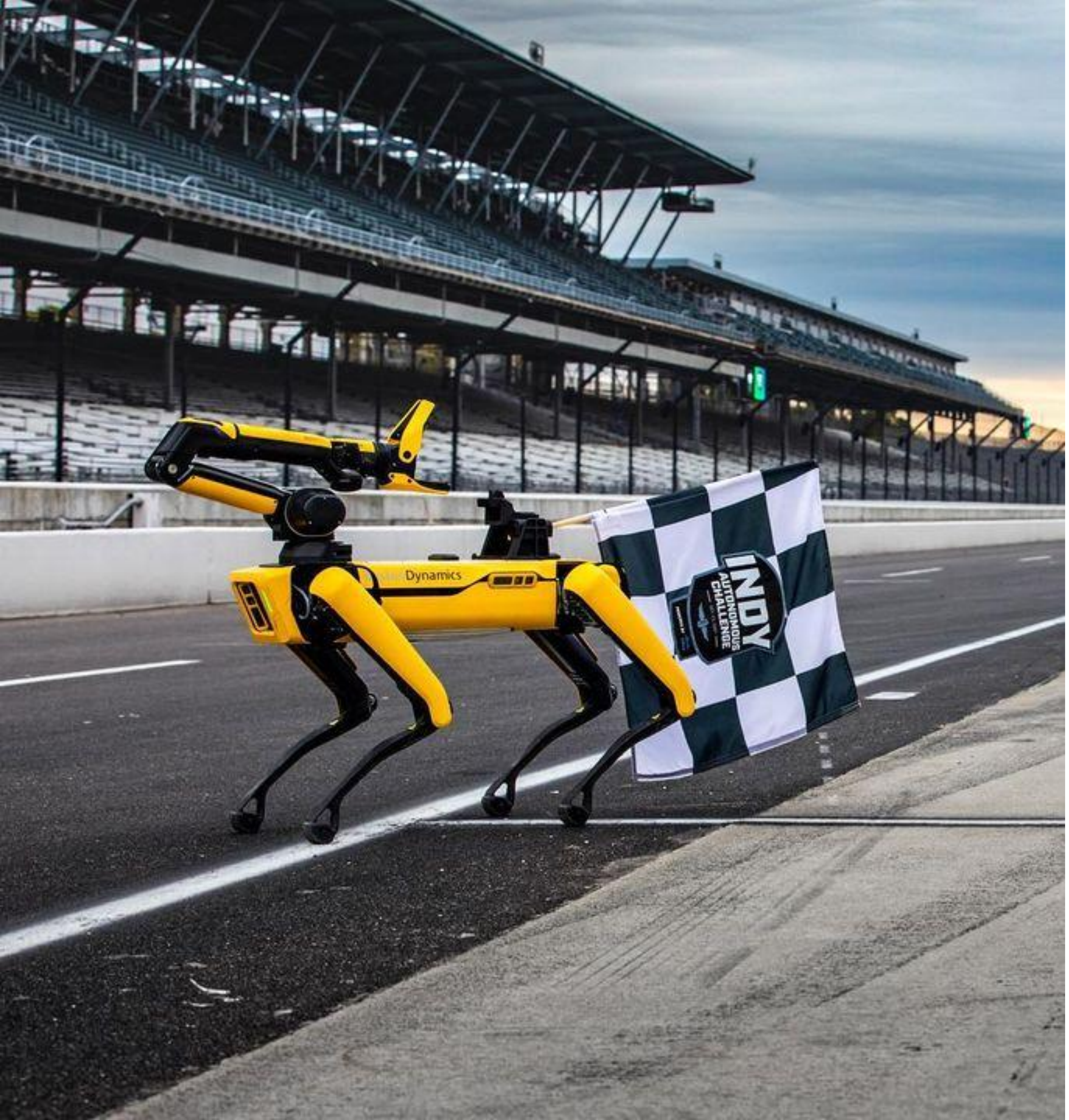
About Me

B. Sc. – TUM Mechanical Engineering (2019)

M. Sc. – TUM Automotive Engineering (2022)

Researcher – TUM FTM (2022 -)

- **IAC@Monza**
- **IAC@CES**
- **EDGAR** 



**Autonomous Racing –
The Future of Racing?**

How can we develop software for an Autonomous Racecar?

How do we improve software for an Autonomous Racecar?

Open Challenges &
Future Efforts

Autonomous Racing – The Future of Racing?

Why Autonomous?

1

Detecting the limits

2

Decision making at the vehicle limits

3

Handling at the vehicle limits

Different Tracks and Conditions

Different vehicle setups

Drivable area detection

High prediction uncertainty

Overtaking at High Speeds

Strategy & Energy Constraints

High speeds, High accelerations

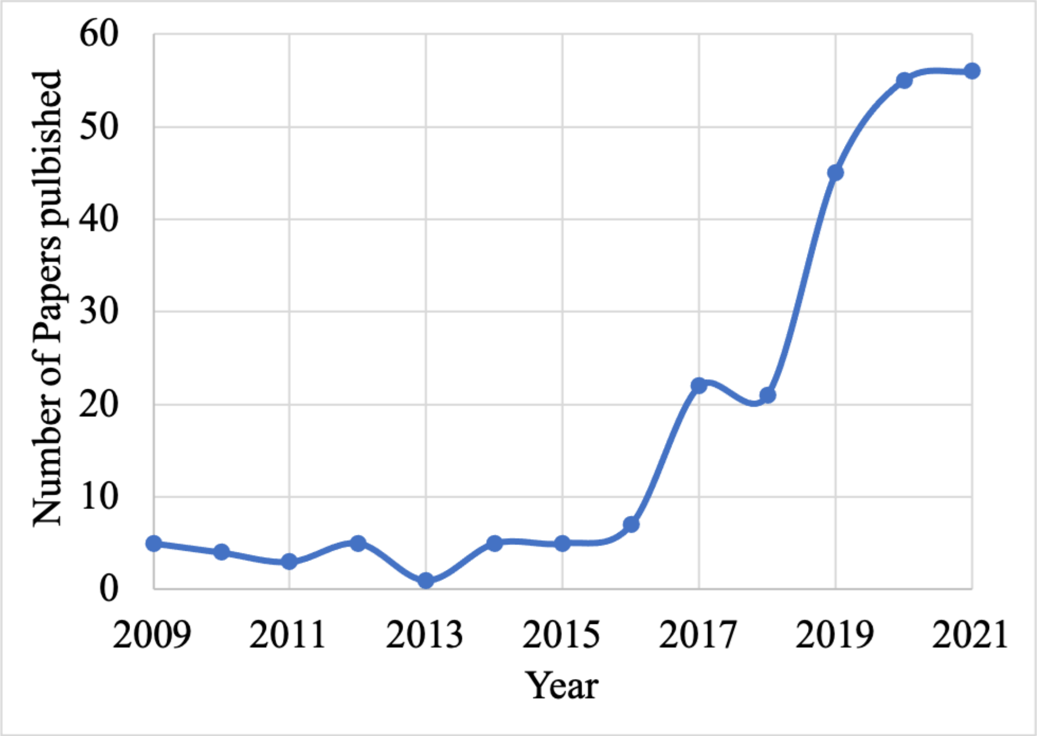
Highly nonlinear behaviour

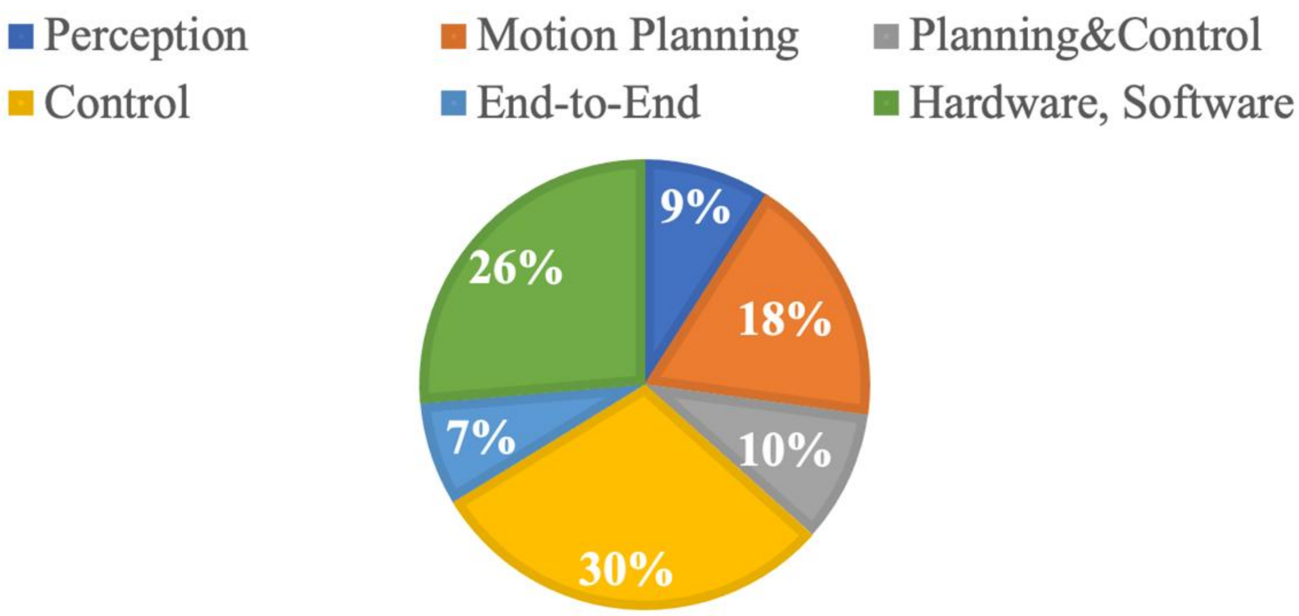
Small reaction times

J. Betz, H. Zheng, A. Liniger, U. Rosolia, P. Karle, M. Behl, V. Krovi, R. Mangharm, “Autonomous Vehicles on the Edge: A Survey on Autonomous Vehicle Racing”, in *Open Journal of Intelligent Transportation Systems (OJ-ITS)*, 2022

Autonomous Racing – The Future of Racing?

The Community





Over 270 scientific Paper in the last 10 Years
Numbers are increasing
Topic becomes more popular

Focus on Software/ Algorithms: Mainly Control
Focus on Hardware and Software Stacks
Little about Simulation & Perception


ROBORACE
TUM
TUM
aws
FERNRIDE
A2RL
Audi piloted driving
IN AJ 354
TU MÜNCHEN
VAIL
VAIL
Audi quattro
Java
CLEMSON
3
Ansys
cisco

Autonomous Racing – The Future of Racing?

Example: The Indy Autonomous Challenge

- Goal: Fully autonomous race
- Over 30 universities worldwide
- $1 million first price

Copyright: Indy Autonomous Challenge

Autonomous Racing – The Future of Racing?

The Indy Autonomous Challenge

Indianapolis 2021
Initital Race

Las Vegas 2022
First Multi Vehicle

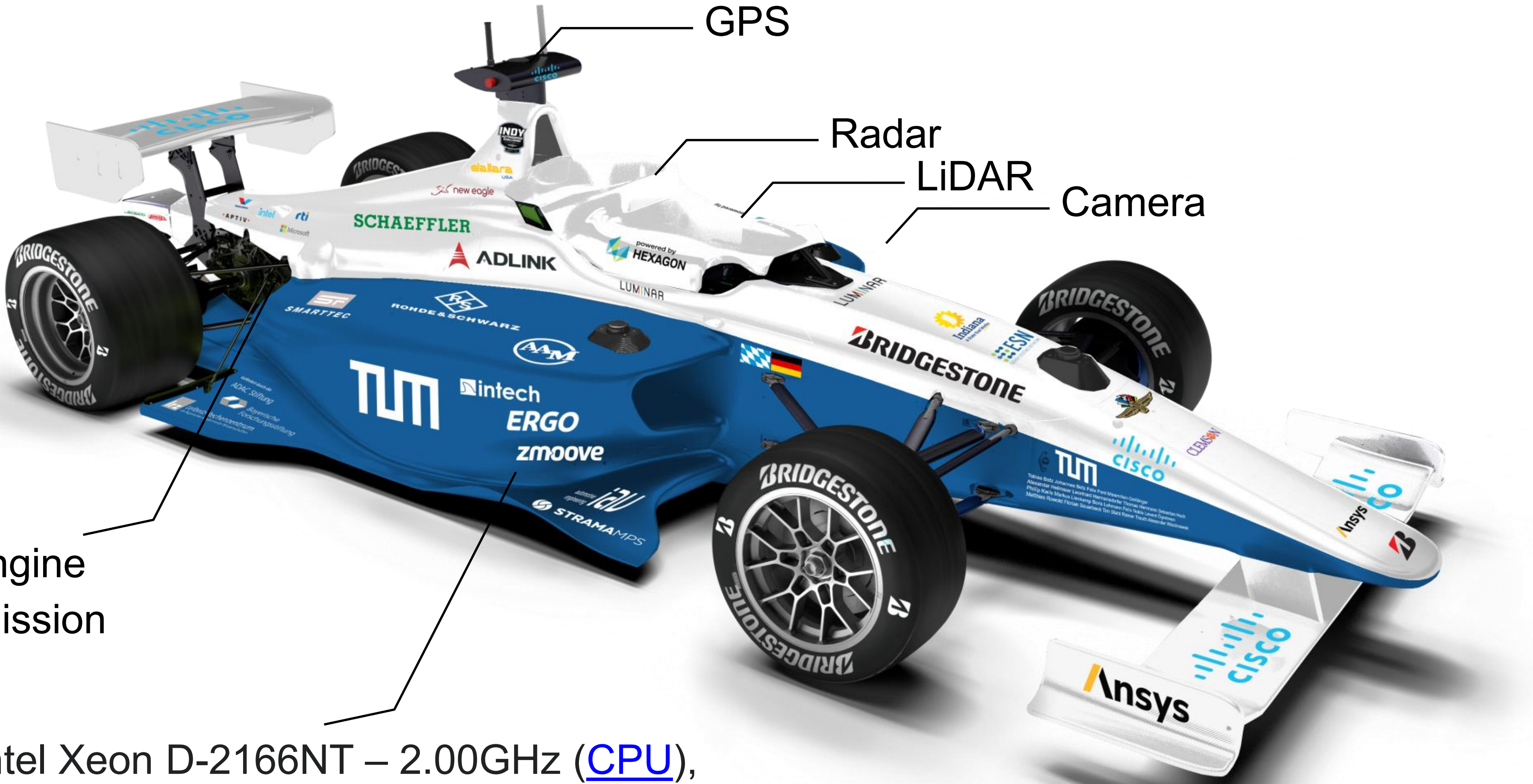
Dallas 2022
Increased Multi Vehicle Complexity

Las Vegas 2023

Monza 2023
First Road course

Autonomous Racing – The Future of Racing?

The Dallara AV-21



GPS

Radar

LiDAR

Camera

Combustion Engine
6 Gear Transmission

Intel Xeon D-2166NT – 2.00GHz (CPU),
NVIDIA RTX A5000 (GPU), 64GB Ram

AUTONOMOUS RACING SERIES – A2RL & IAC

Abu Dhabi Autonomous Racing League (A2RL)

Abu Dhabi Autonomous Racing League

First Race: April 2024

Base Vehicle: Dallara SF-23

Top Speed: 250 km/h

Website: https://a2rl.io/

Indy Autonomous Challenge (IAC)

First Race: October 2021

Base Vehicle: Dallara AV-24

Top Speed: 280 km/h

Website: www.indyautonomouschallenge.com

Autonomous Racing – The Future of Racing?

A Definition

Autonomous Racing is a branch of autonomous driving research conducted by **international researchers** with the goal to drive at **high speeds** in an adversarial **multi-vehicle environment** in **simulation** and **real-world** at **international competitions**.

Autonomous Racing –
The Future of Racing?

How can we develop software for an Autonomous Racecar?

How do we improve software for an Autonomous Racecar?

Open Challenges & Future Efforts

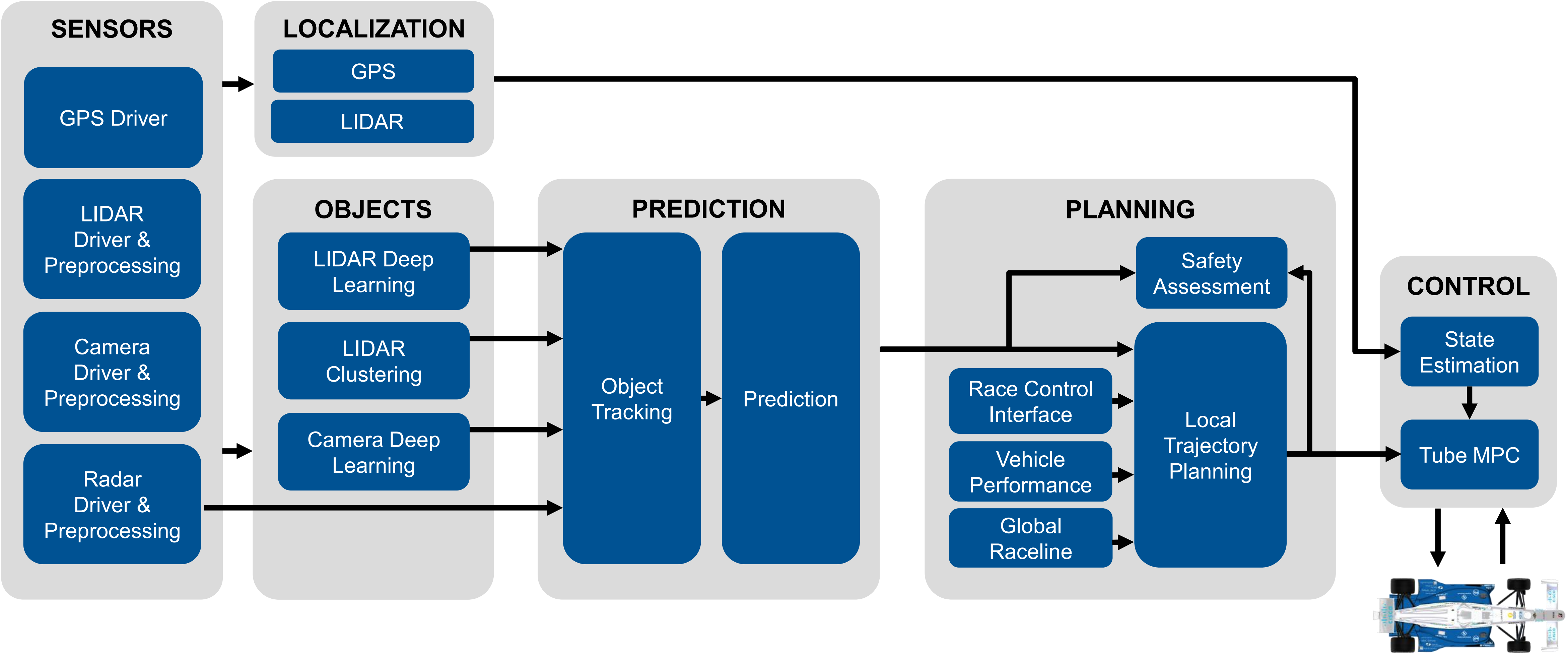
Tools and Middleware

- **ROS2** is used as a main middleware to exchange information between software modules
- **Modular software design** based to leverage multi-core CPU and fast development times
- **Containerization** of software modules for deployment, versioning and dependency isolation

- **Python** for tooling and **machine learning** training
- **ONNX** and **TensorRT** for machine learning inference
- **C++ for performance critical software components**

The Software Architecture

Autonomous Racing



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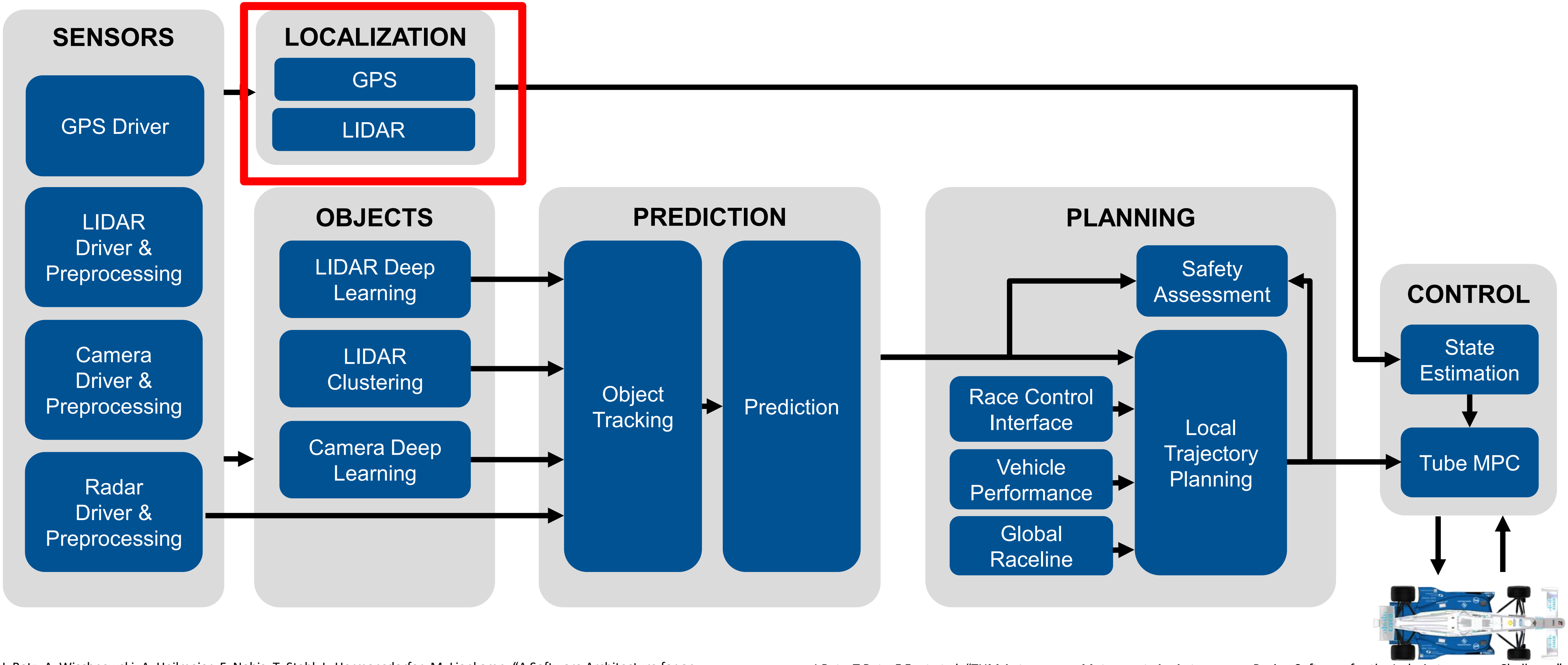
The Software Architecture

Autonomous Racing



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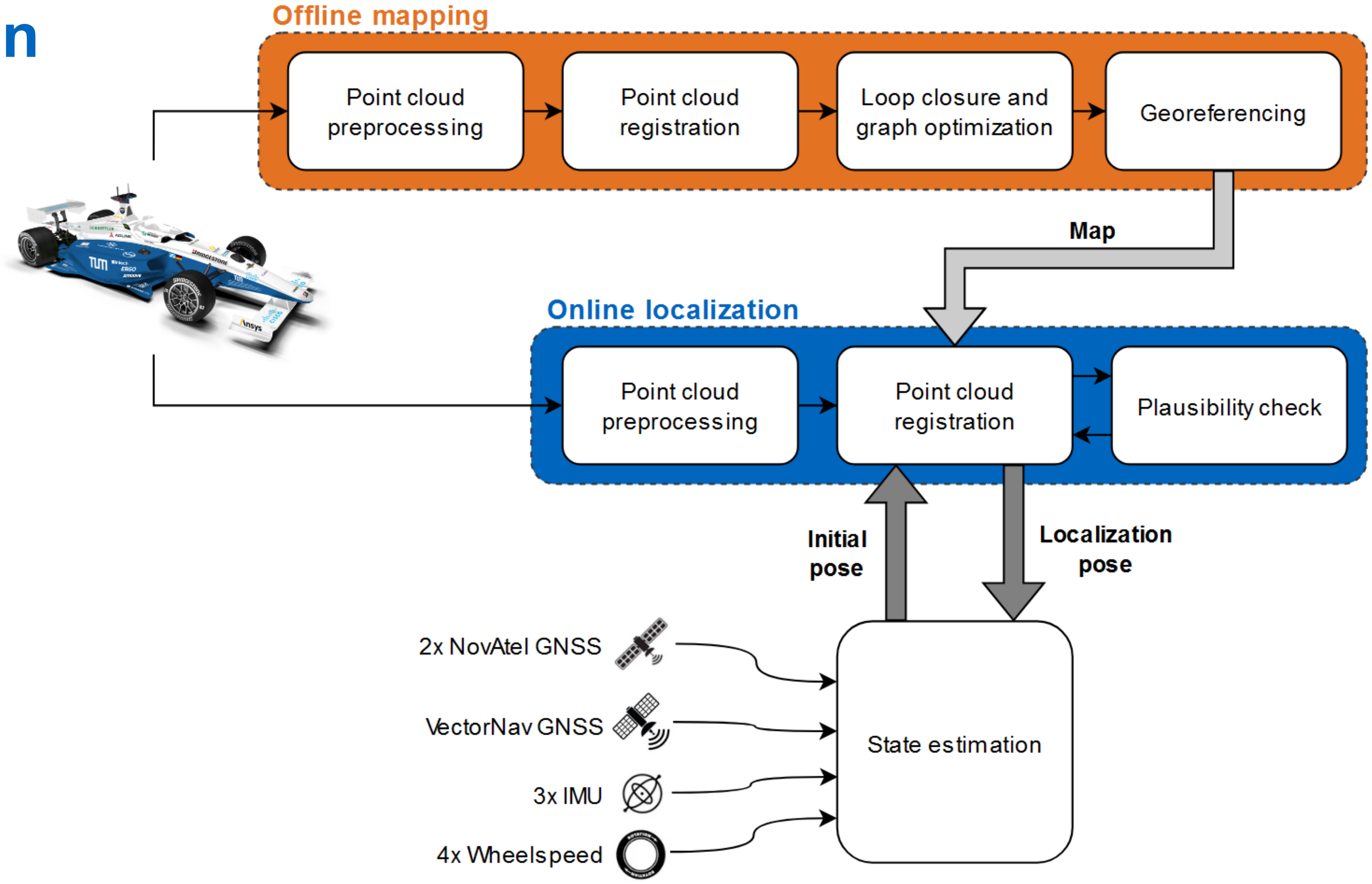
Localization

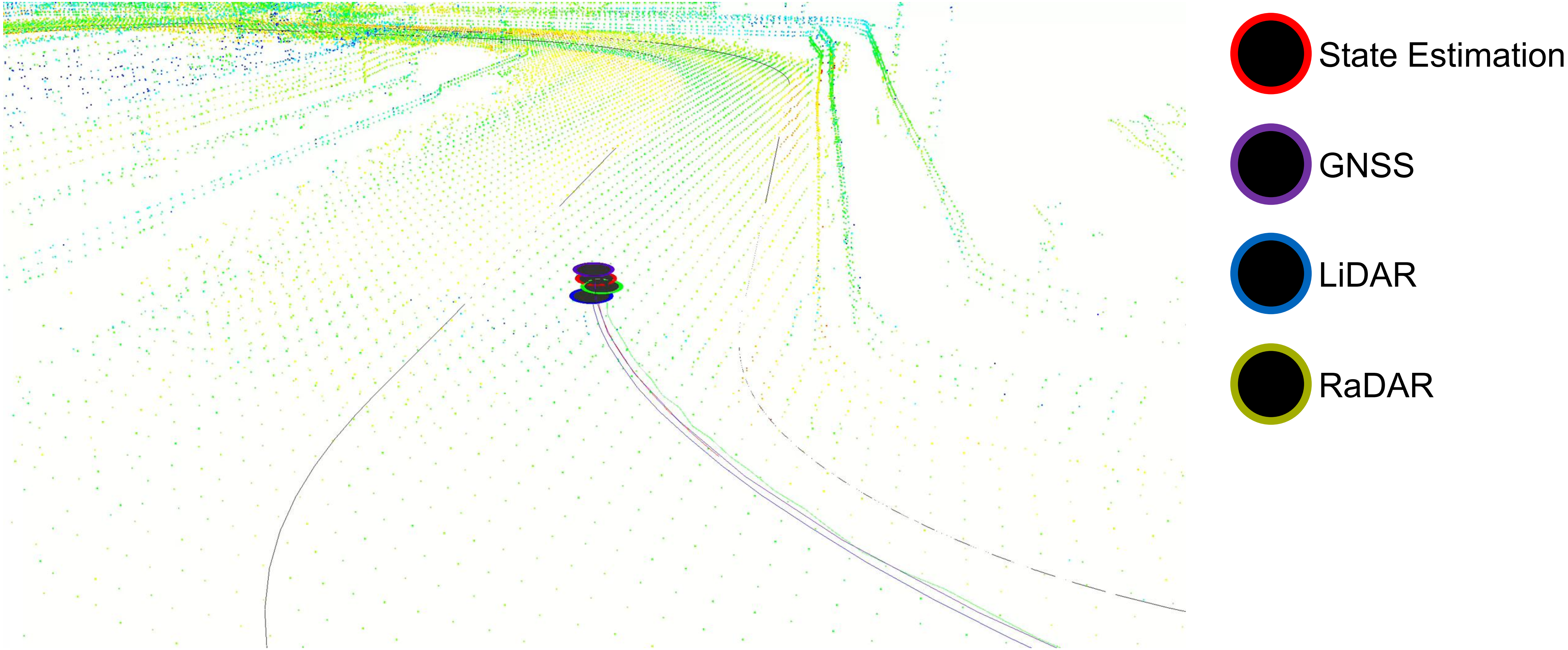
Multi Sensor Fusion



A. Wischnewski, T. Stahl, J. Betz, B. Lohmann, “Vehicle Dynamics State Estimation and Localization for High Performance Race Cars“ in *IFAC-PapersOnLine*, vol. 52, no. 8, p. 154–161, 2019, doi: 10.1016/j.ifacol.2019.08.064

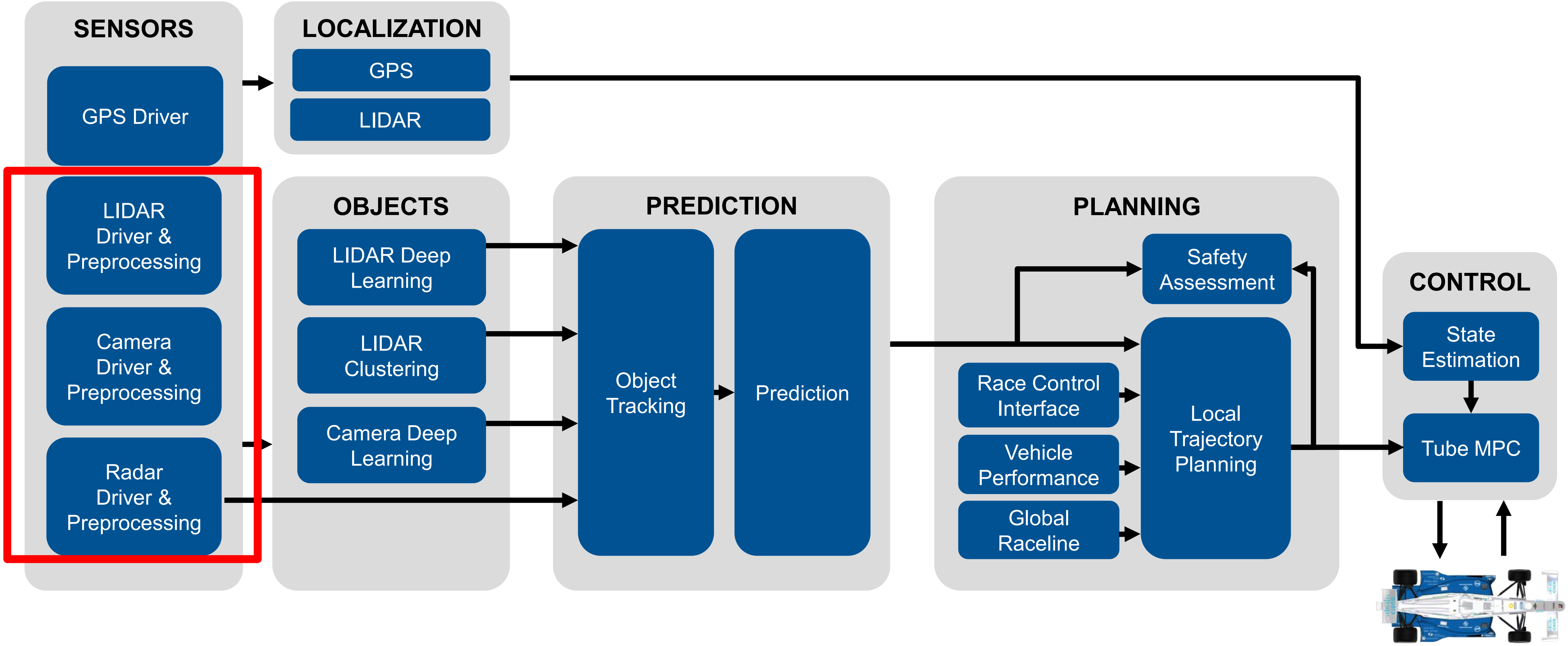
F. Sauerbeck, L. Baierlein, J. Betz, M. Lienkamp, “A Combined LiDAR-Camera Localization for Autonomous Race Cars” in *SAE International Journal of Connected and Automated Vehicles*, vol. 5, no. 1, Jan. 2022, doi: 10.4271/12-05-01-0006

Localization
Multi Sensor Fusion



The Software Architecture

Autonomous Racing



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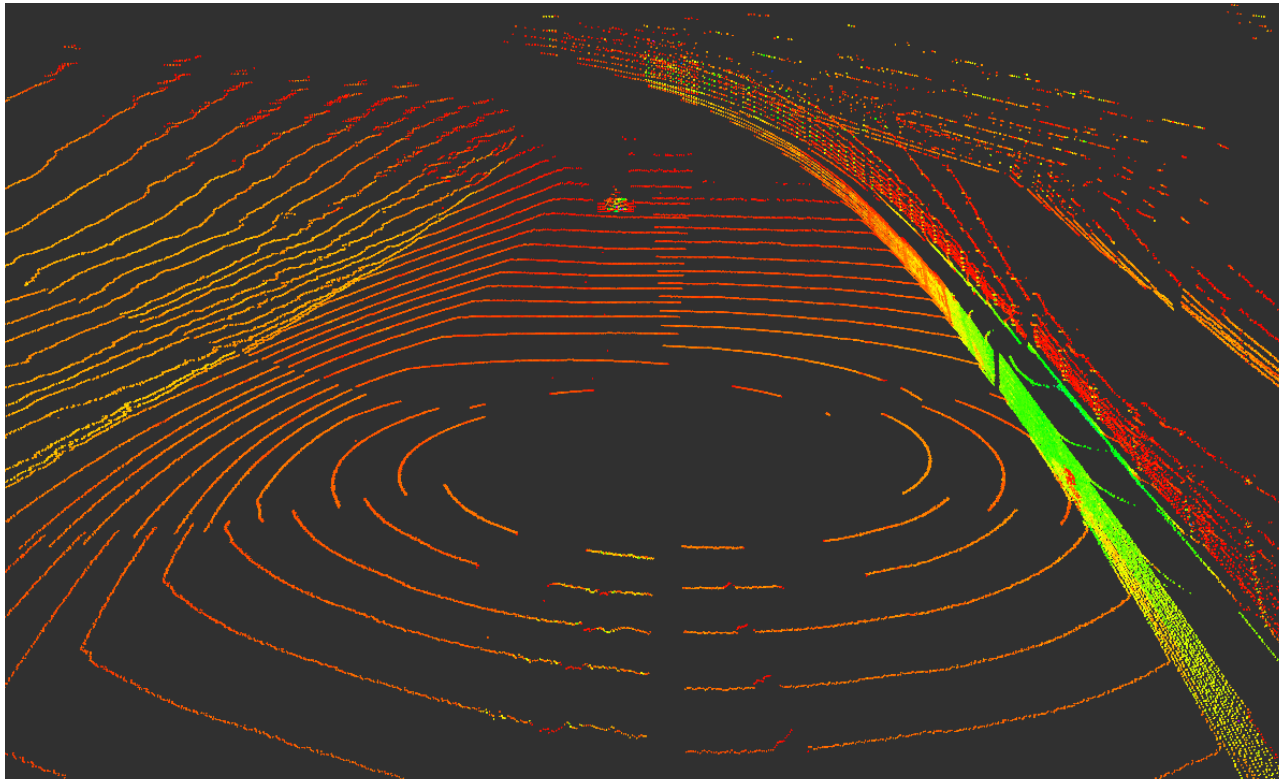
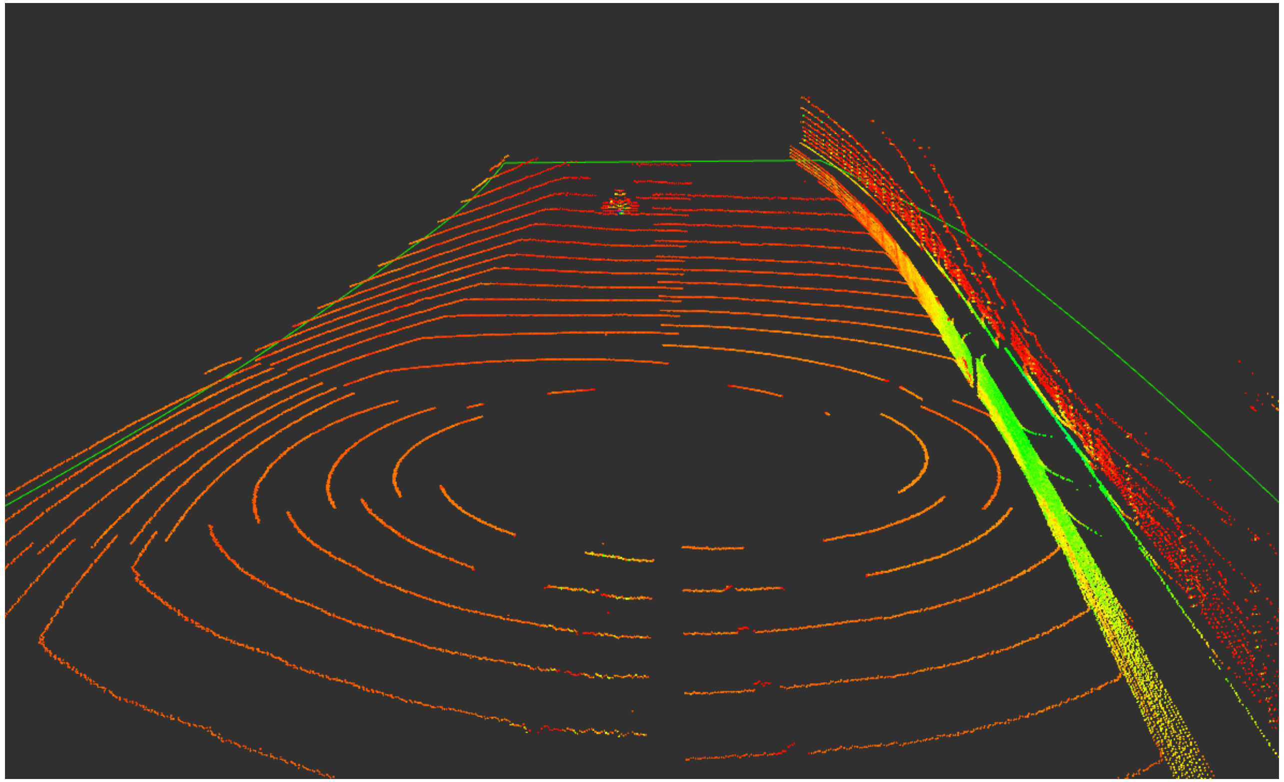
Object Detection – Preprocessing



- Most points not relevant for driving task
- Efficient preprocessing to reduce computation time in succeeding algorithms
- Geometry + Dust Filter
- Ground Filter
- Wall Filter

Object Detection – Preprocessing



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- **Geometry + Dust Filter**
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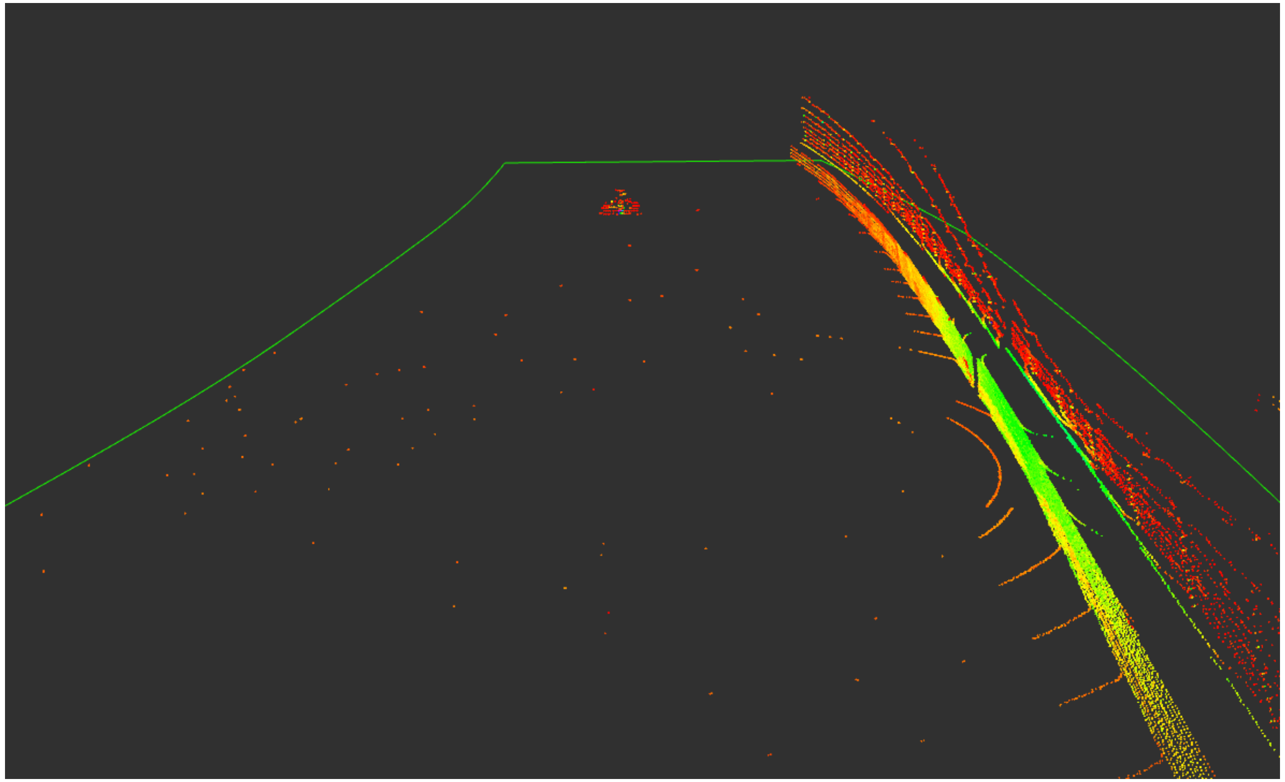
Object Detection – Preprocessing



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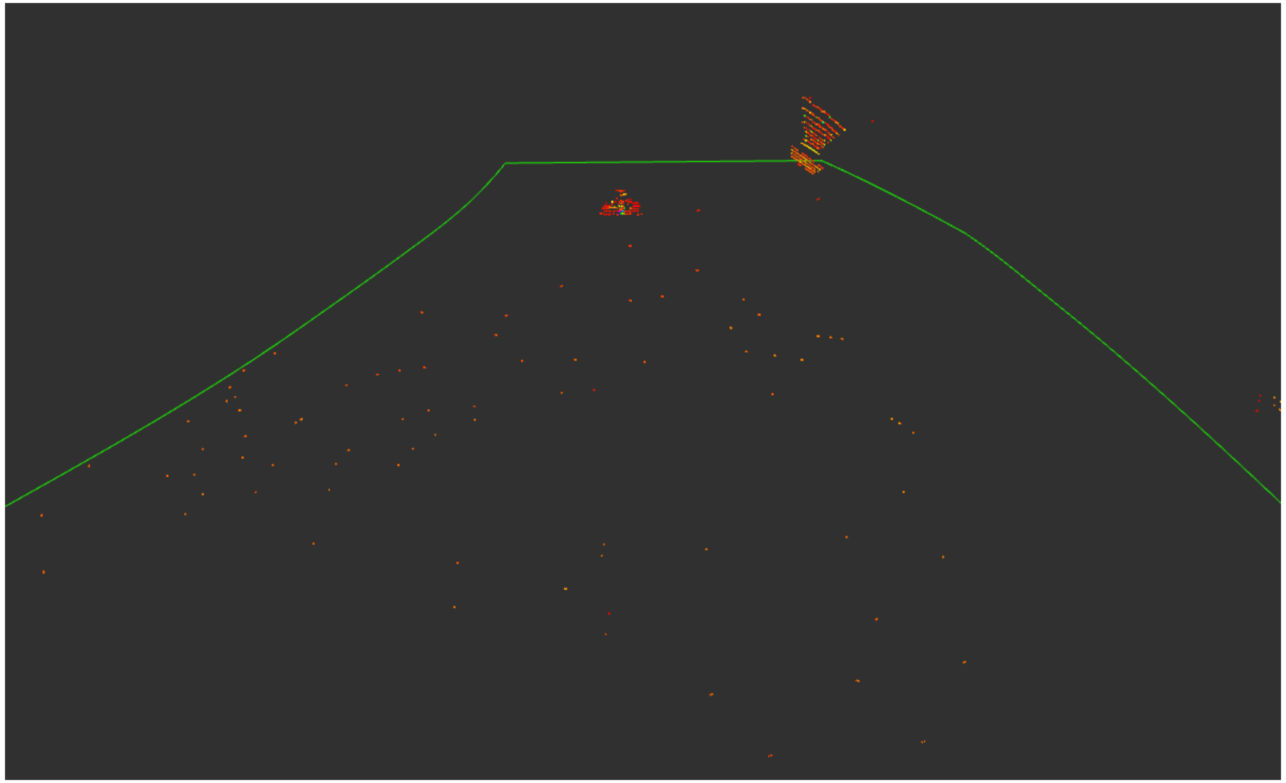
Object Detection – Preprocessing

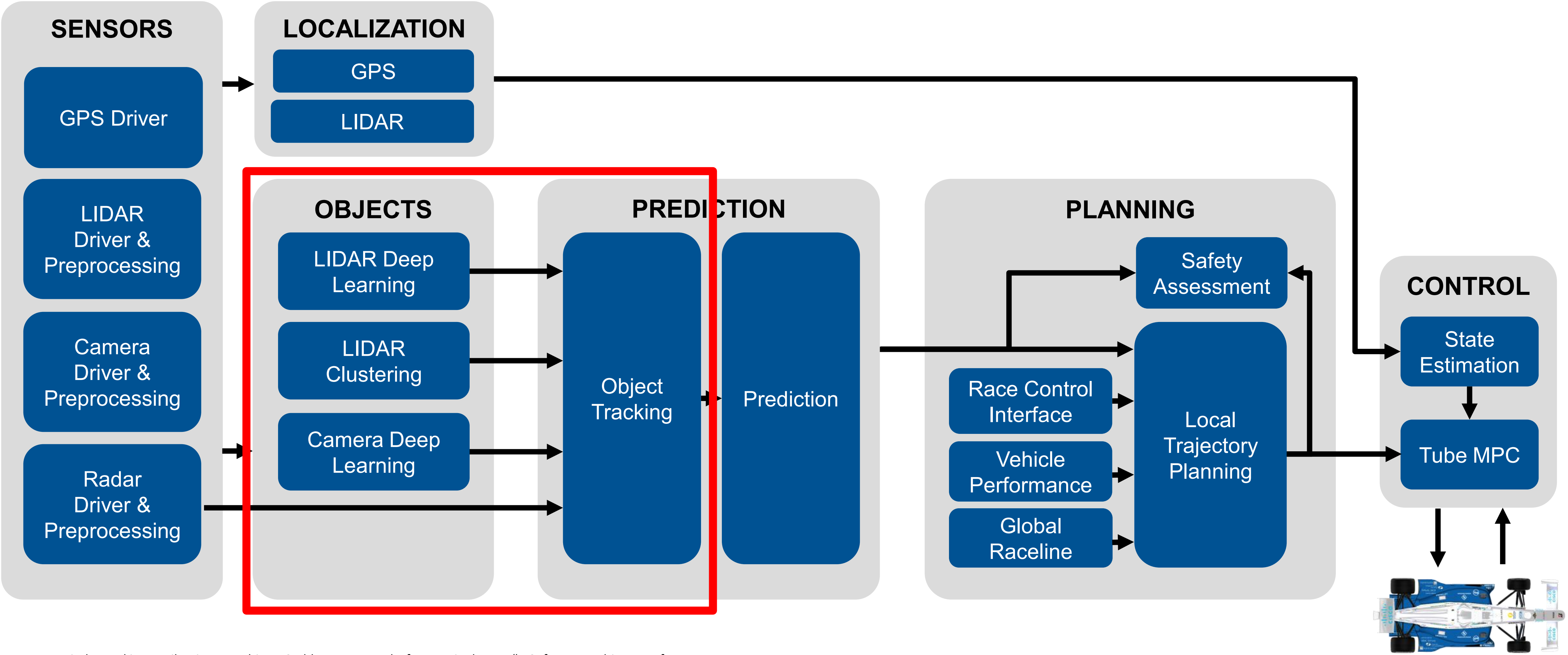


- Most points not relevant for driving task
- Efficient preprocessing to reduce computation time in succeeding algorithms
- Geometry + Dust Filter
- Ground Filter
- **Wall Filter**

→ **Total time: ~25-30ms**

The Software Architecture

Autonomous Racing



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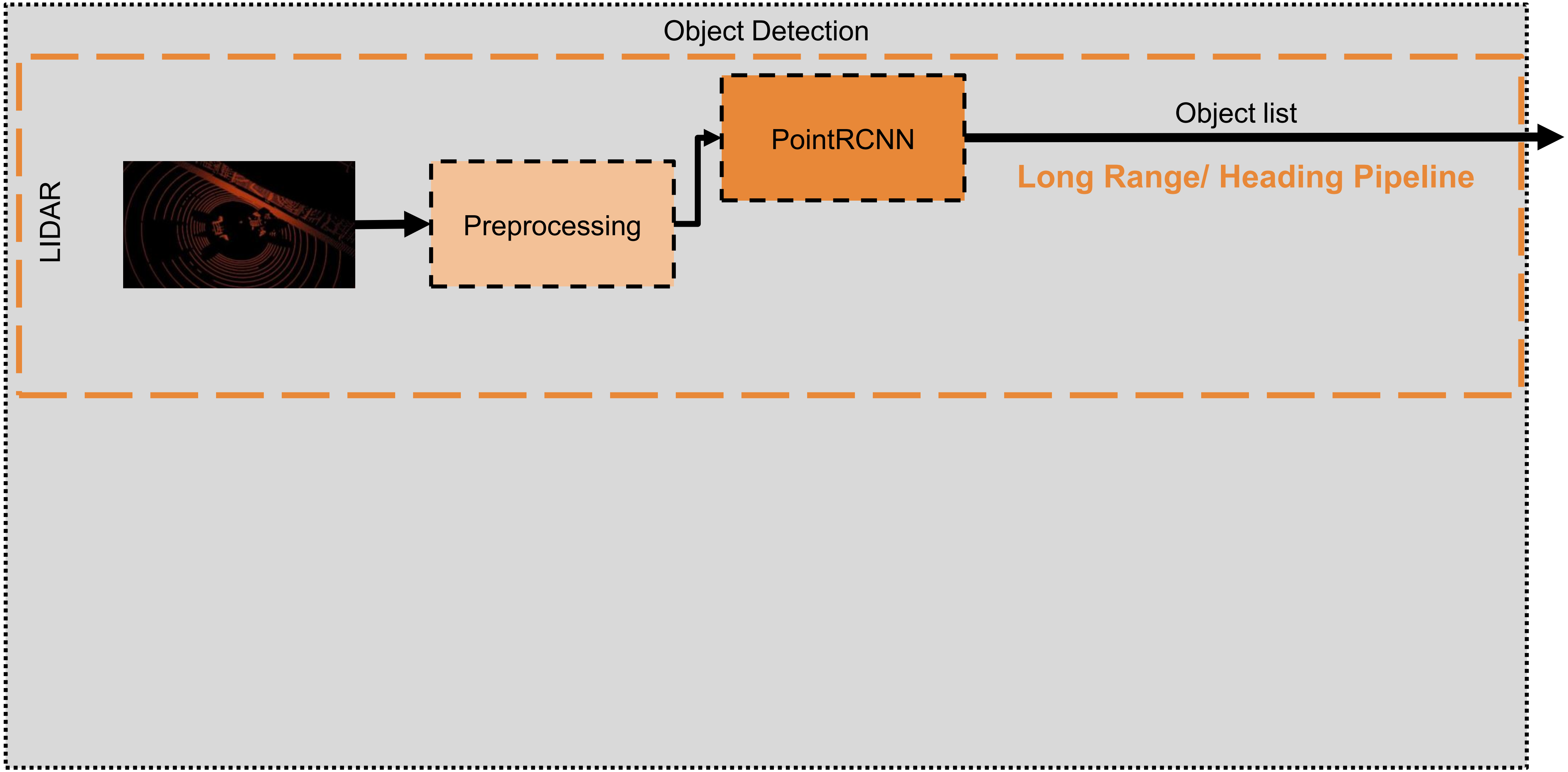
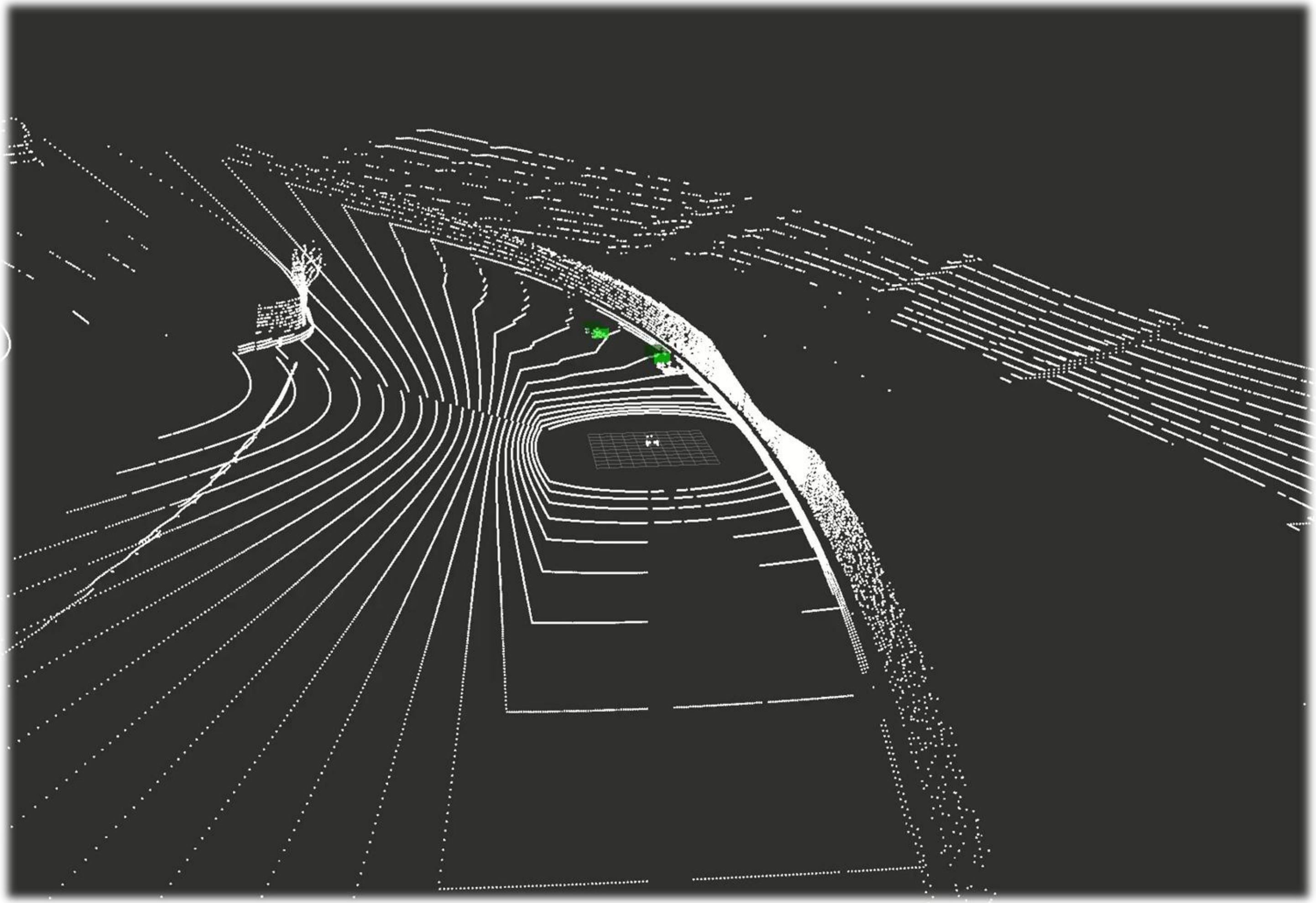
Object Detection



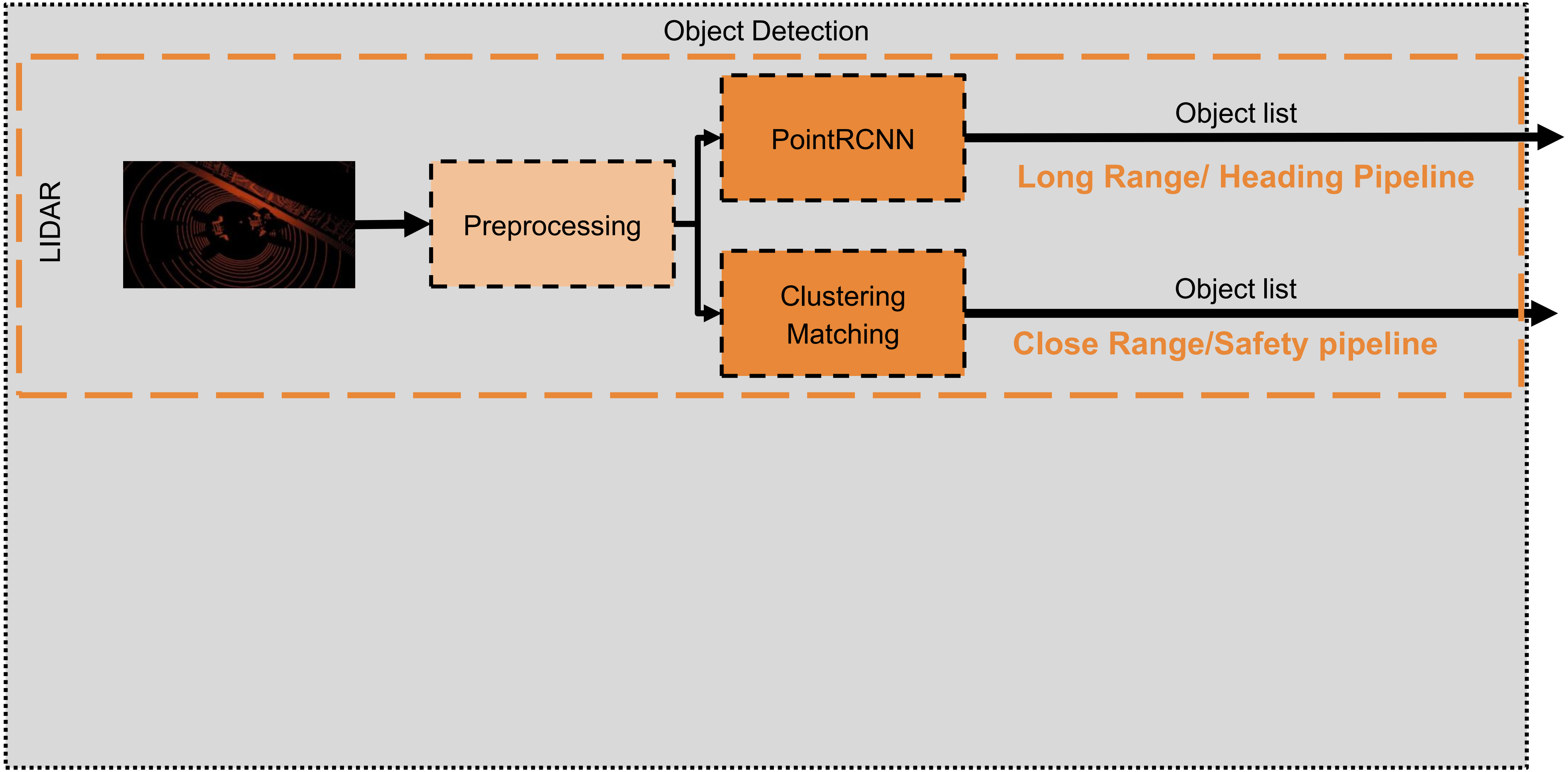
Object Detection – LiDAR Deep Learning



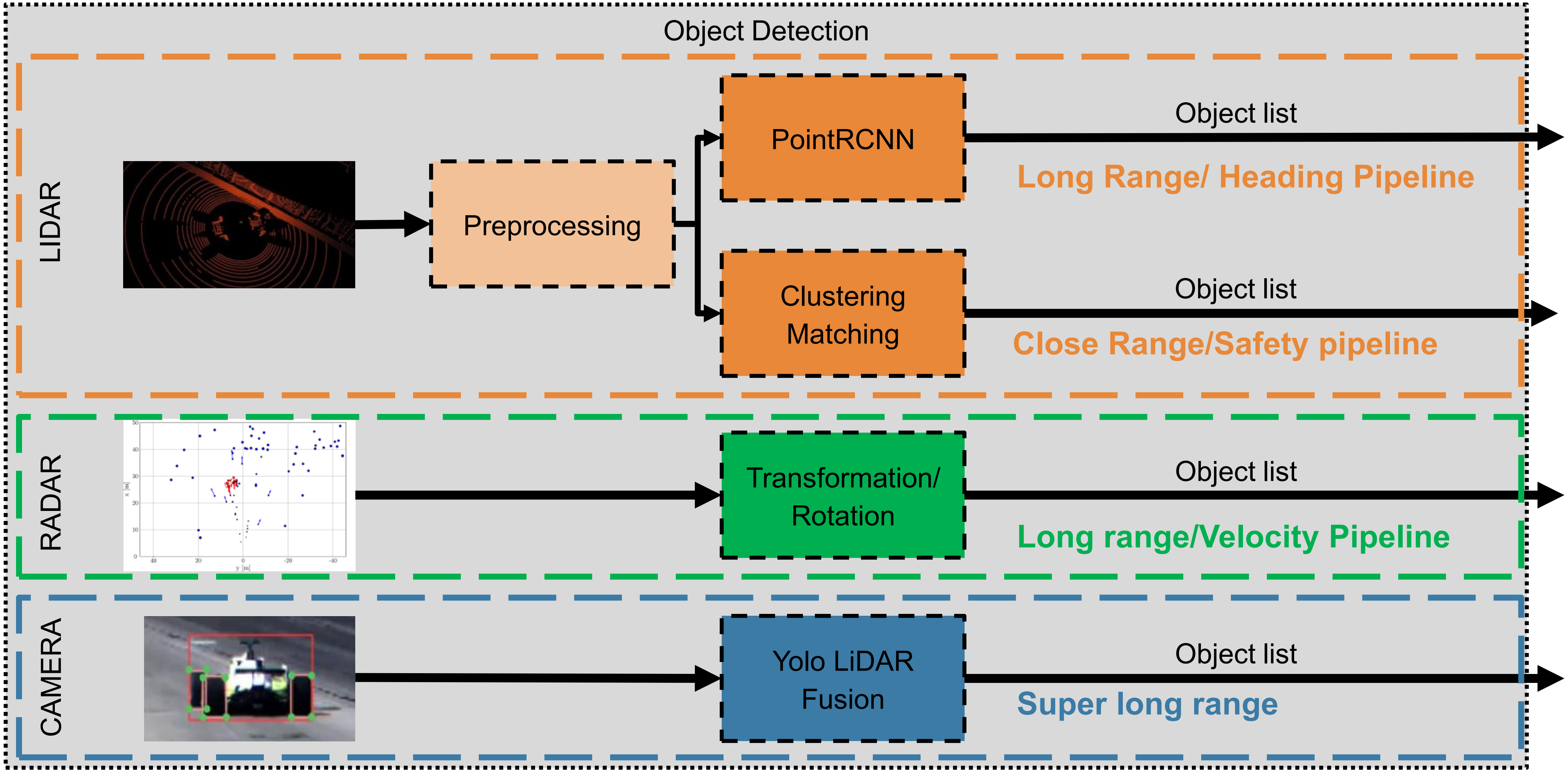
PointRCNN

- Based on PointNet
- Two-stage detector
- 3D Detection
- **Labeled training data required**

Object Detection

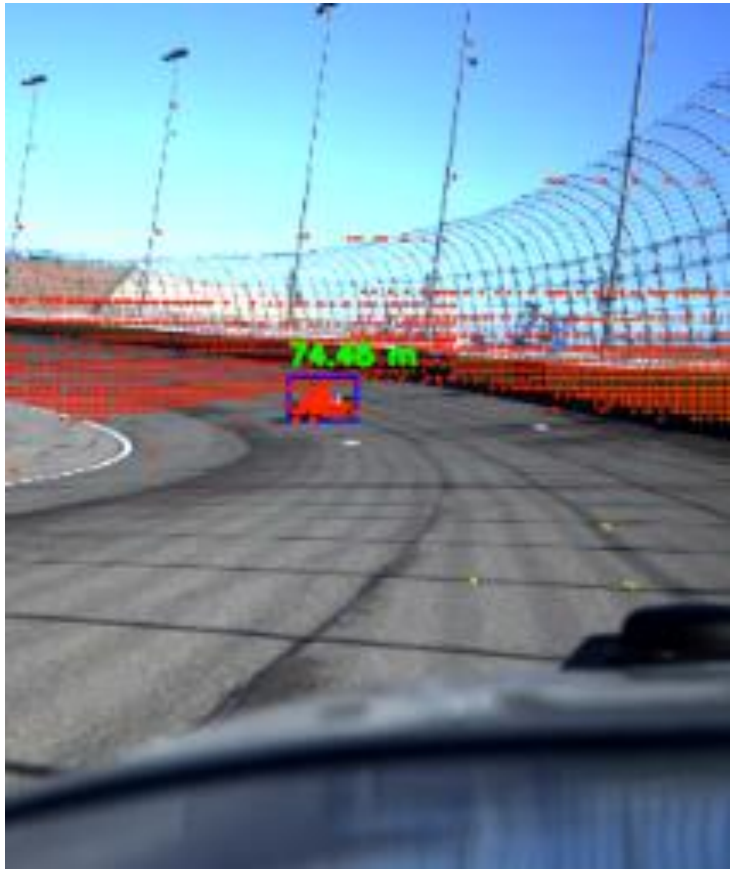


Object Detection



Object Detection

Camera Deep Learning



- Combines camera image with LiDAR pointcloud
- Fast and simple detection in frustrum
- >120m detection
- **Camera/LiDAR calibration and synchronization necessary**

Object Detection

Late Fusion and Tracking

Object Fusion and Tracking

1. **Transformation**
   Local → Global Coordinate System

2. **Plausibility Check**
   Merge of multiple tracks
   Out-of-Track-Filter

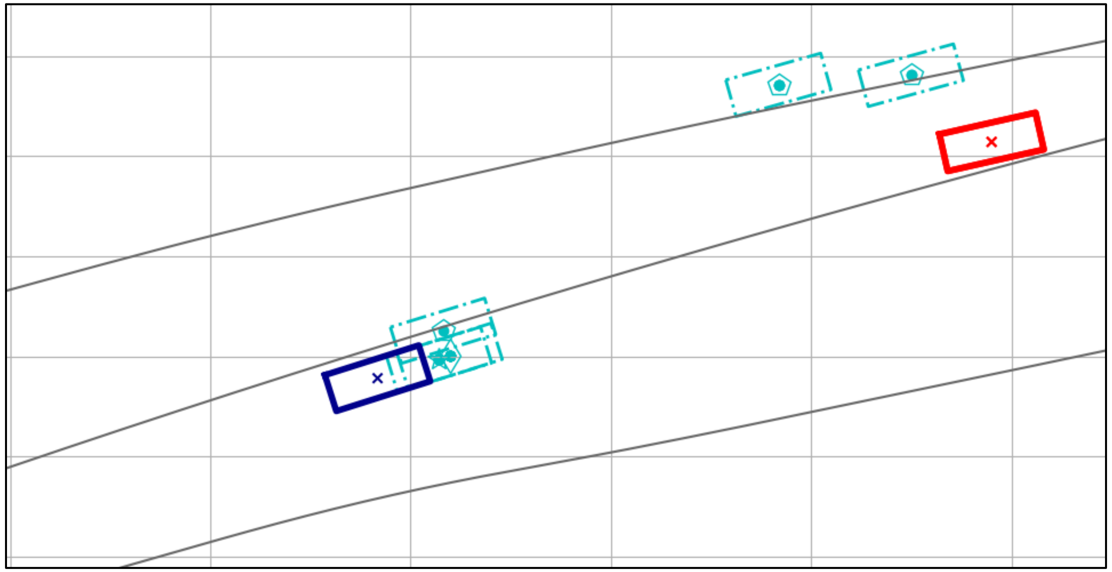
3. **Object Matching**
   Hungarian Method

4. **State Estimation**
   Extended Kalman Filter

5. **Delay Compensation**
   Backware-Forward-Integration

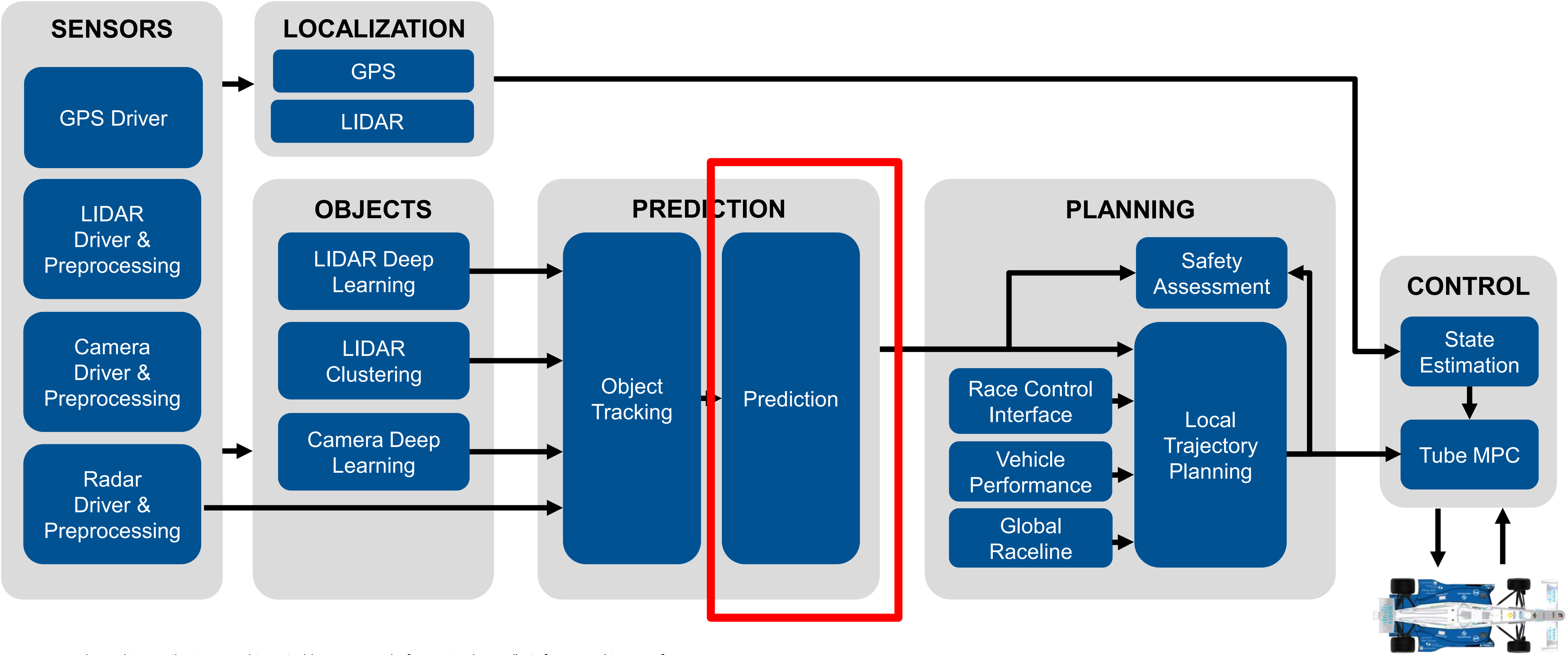
→ Unified Object list

Object Matching



The Software Architecture

Autonomous Racing



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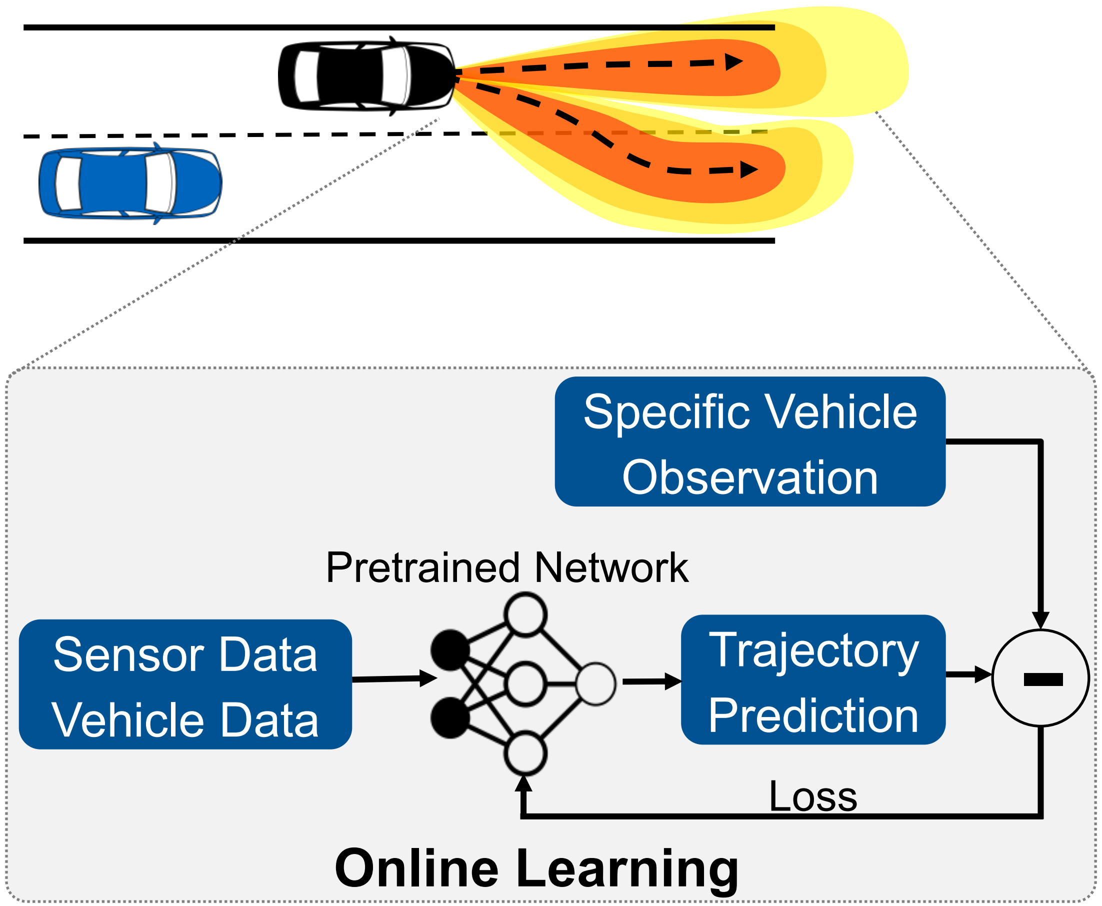
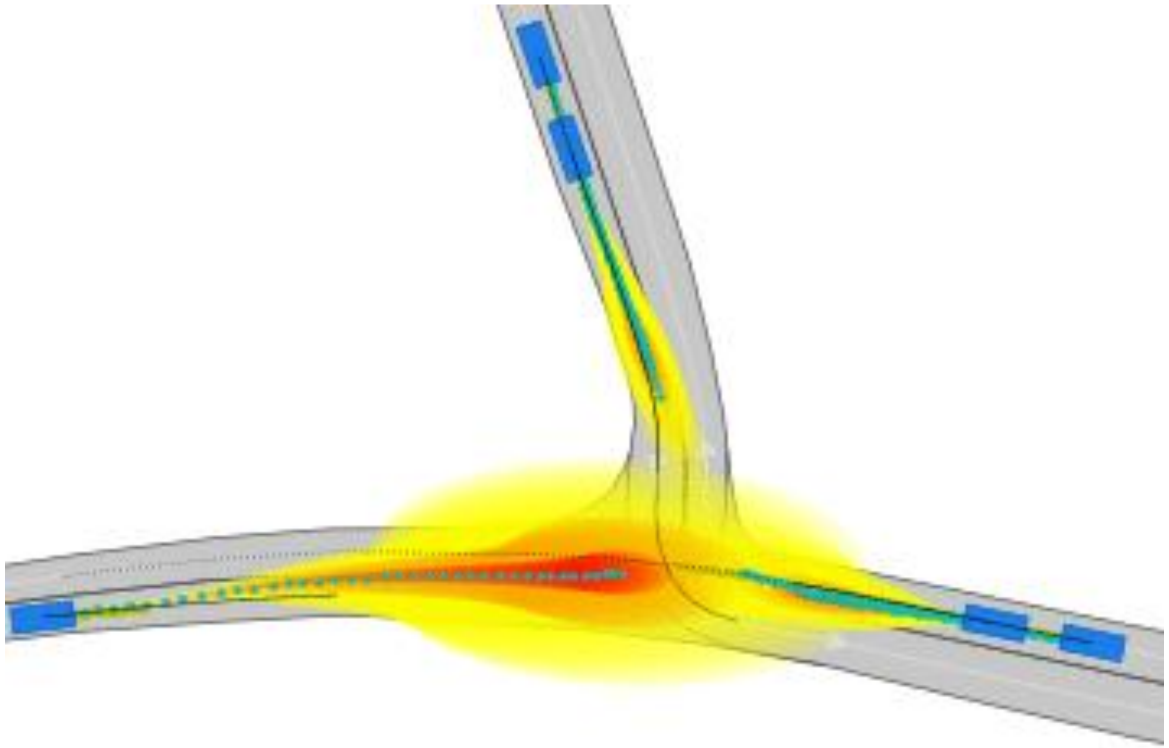
Vehicle Motion Prediction
Online Learning





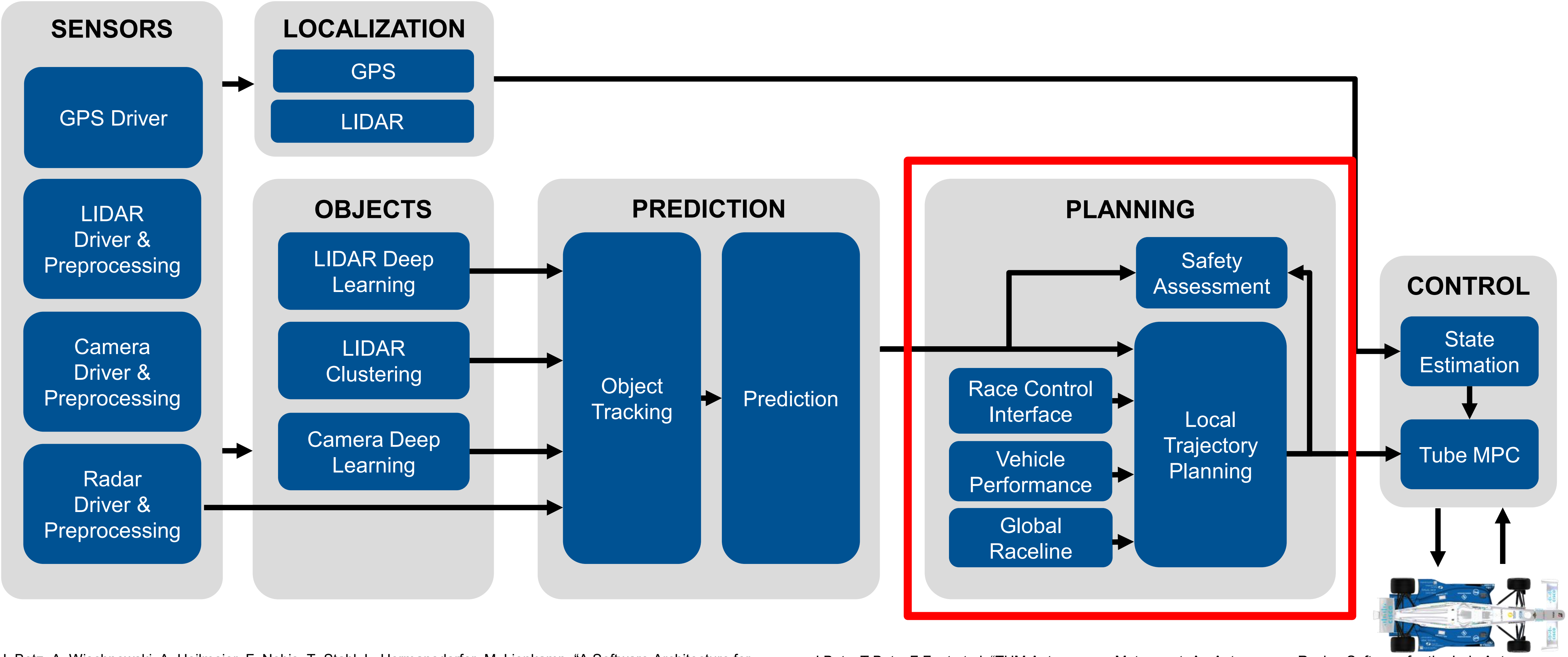
- Why? **Predict correct behavior** of other vehicles
- Creation of **online learning and adaption** network
- Use **observations during inference**
- **Reduce** predicted uncertainty of predicted trajectories

M. Geisslinger, P. Karle, J. Betz, M. Lienkamp, "Watch-and-Learn-Net: Self-supervised Online Learning for Probabilistic Vehicle Trajectory Prediction" in *IEEE Inter. Conference on Systems, Man, and Cybernetics (SMC),* 2021

P . Karle, M. Geisslinger, J. Betz, M. Lienkamp, "Scenario Understanding and Motion Prediction for Autonomous Vehicles – Review and Comparison" in IEEE ITS Transactions, *in Print*

The Software Architecture

Autonomous Racing



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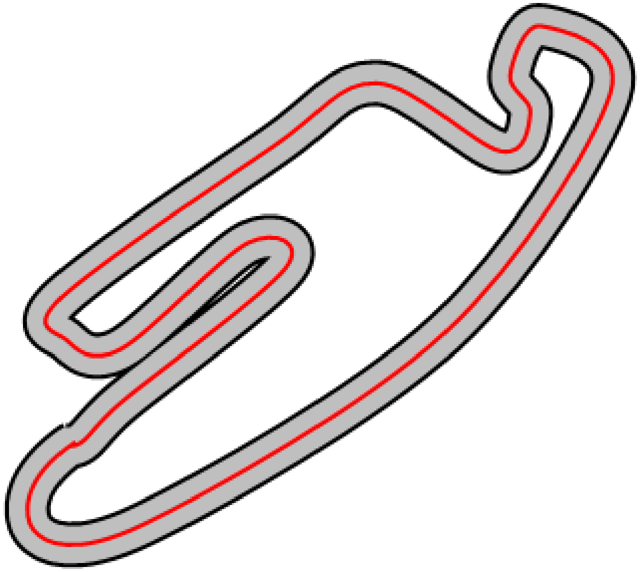
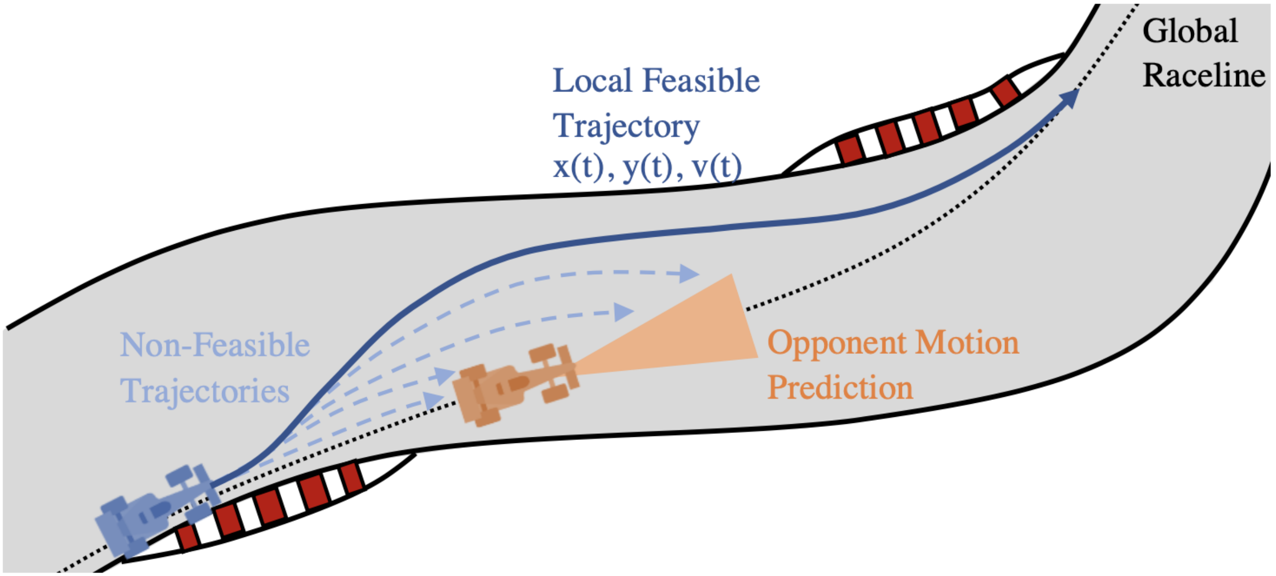
Planning

Global vs. Local Planning





- **Static** environment
- Plan a trajectory **around a race track**
- **No moving objects** on the track
- Plan **fastest** trajectory
- Primary **Offline** – single execution

- **Dynamic** environment: Race, Follow, overtake
- Plan a trajectory, starting at the ego-vehicle and ending at the planning horizon
- Plan **safe** trajectory
- **Online only** – real-time re-calculation

A. Heilmeier, A. Wischnewski, L. Hermansdorfer, J. Betz, M. Lienkamp, B. Lohmann, "Minimum curvature trajectory planning and control for an autonomous race car" in *Vehicle System Dynamics*, vol. 58, no. 10, p. 1497–1527, Jun. 2019 doi: 10.1080/00423114.2019.1631455

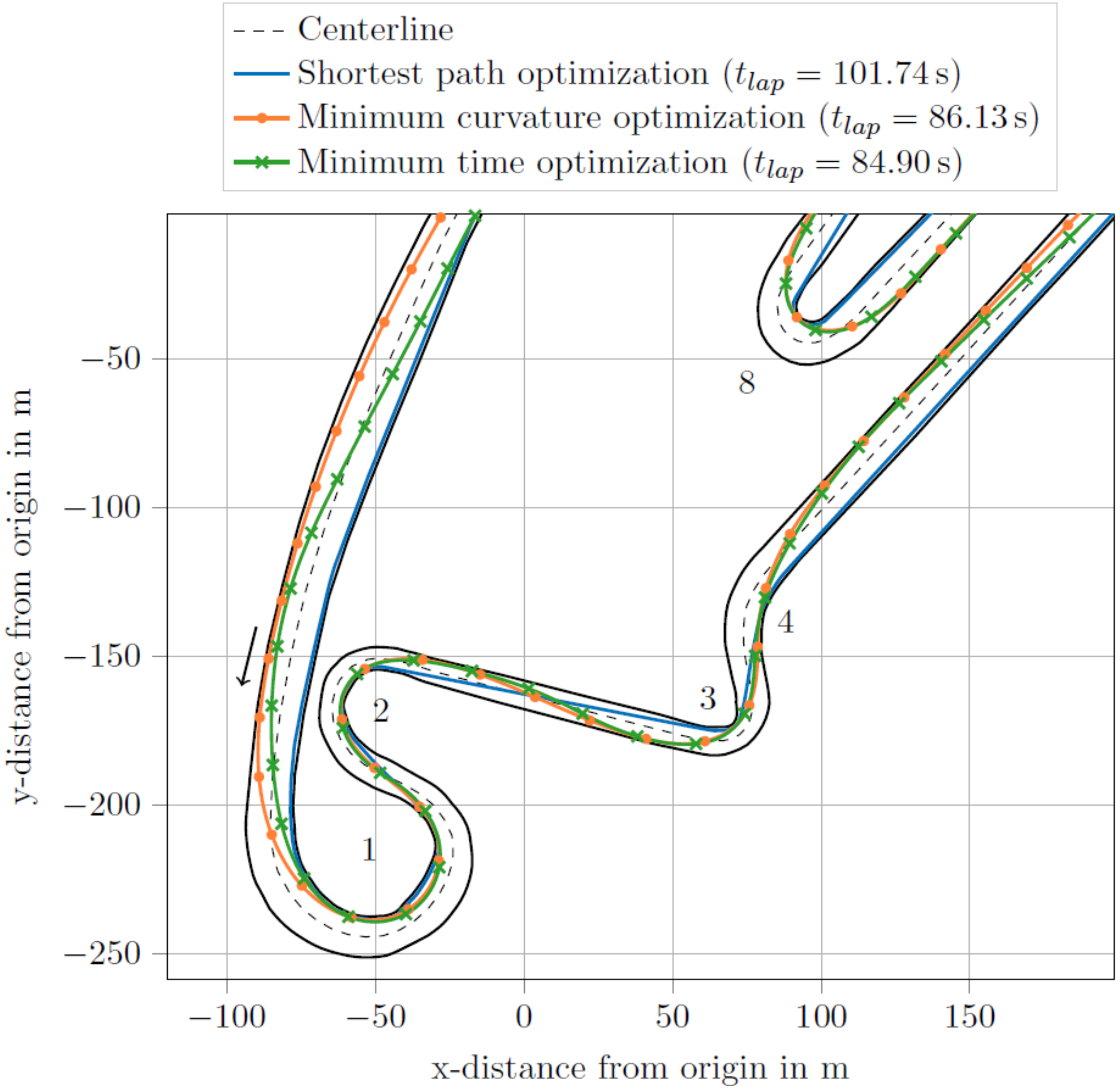
F. Christ, A. Wischnewski, A. Heilmeier, and B. Lohmann, "Time-optimal trajectory planning for a race car considering variable tyre-road friction coefficients," Vehicle System Dynamics, vol. 59, no. 4

Planning

Global Planning

Optimization Approach:

- Different possibilities to create an optimal path around the racetrack
- Global lap time minimization based on **optimal control**
- **NLP:** Nonlinear problem optimization
- Vehicle performance integrated as single or **double track model**
- **Knowledge about available Grip** can be integrated with an additional map
- **Runtime:** 5-10 minutes



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Motion planning in complex scenarios

Local Planning

- Non-convex problem due to multiple agents
- Trade-off between global optimum and runtime
- Required to work along the full vehicle dynamics range

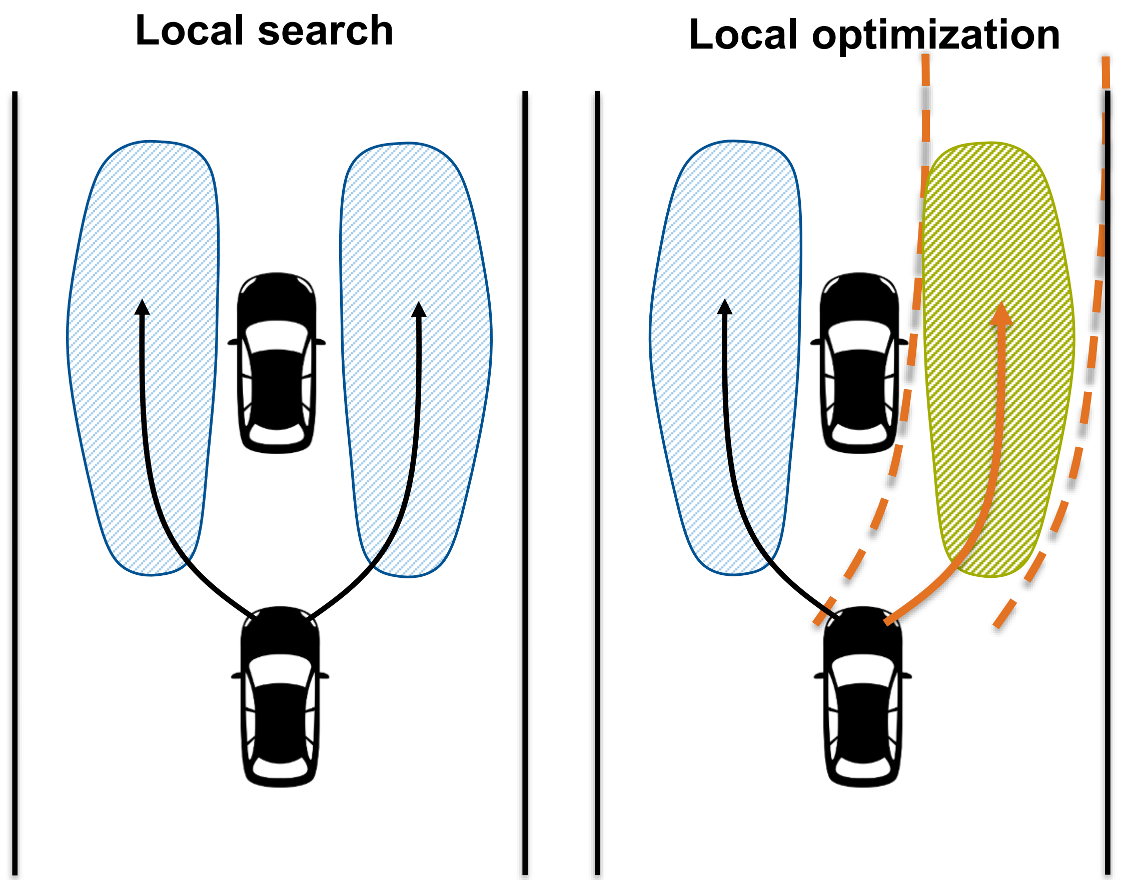
Sampling based search → Re-optimzation with MPC

✓ Efficient implementation

✓ Arbitrary cost functions

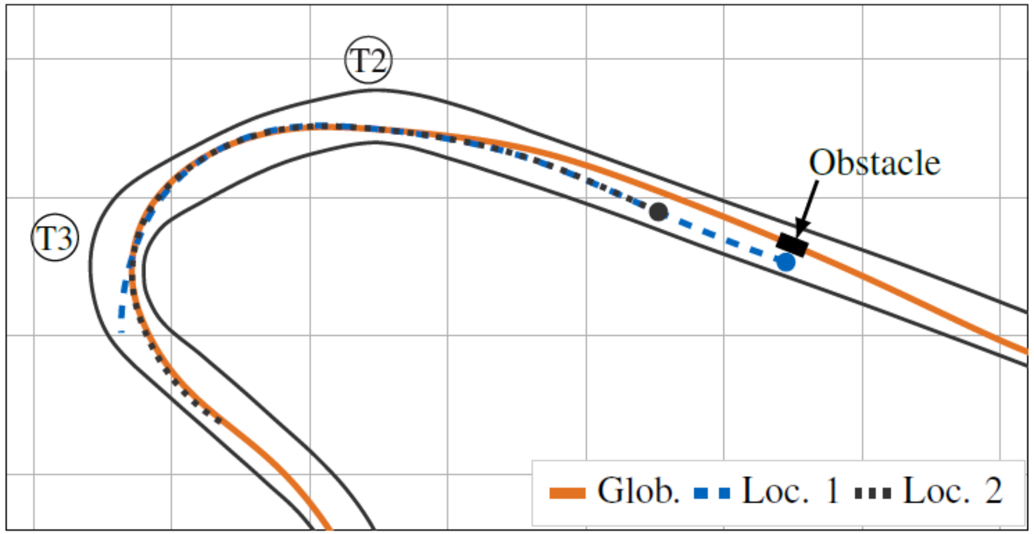
✓ Refine solution up to centimeter level

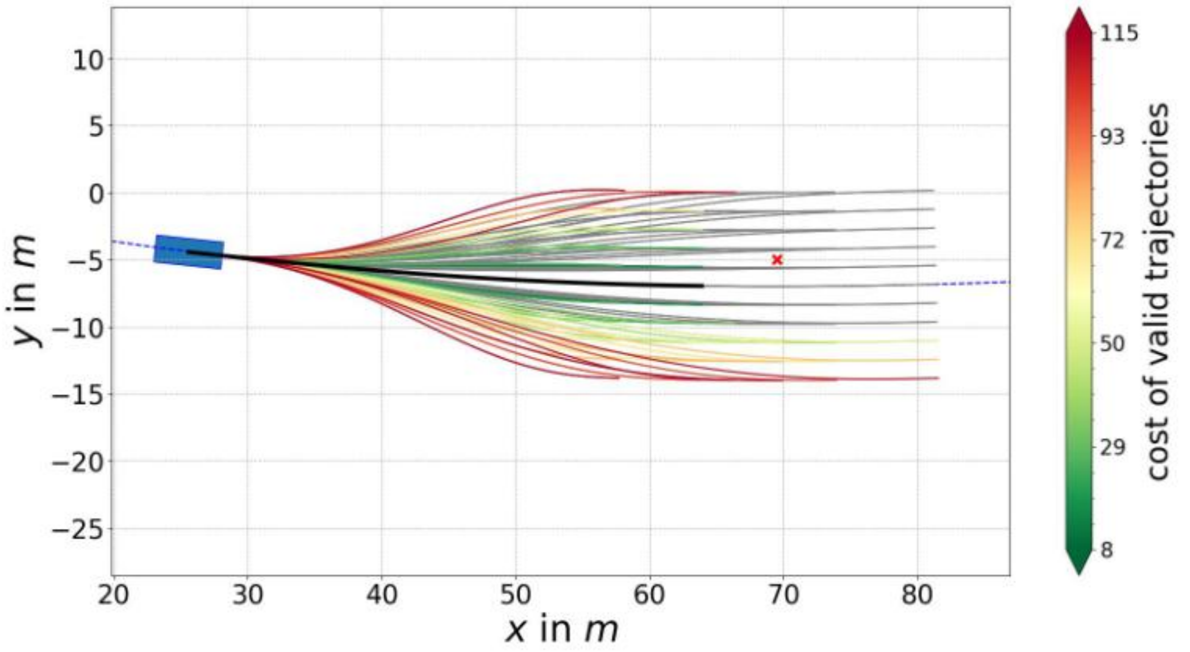
✓ Handles disturbances and uncertainties



Motion Planning in Complex Scenarios

Local Planning – Sampling Planner

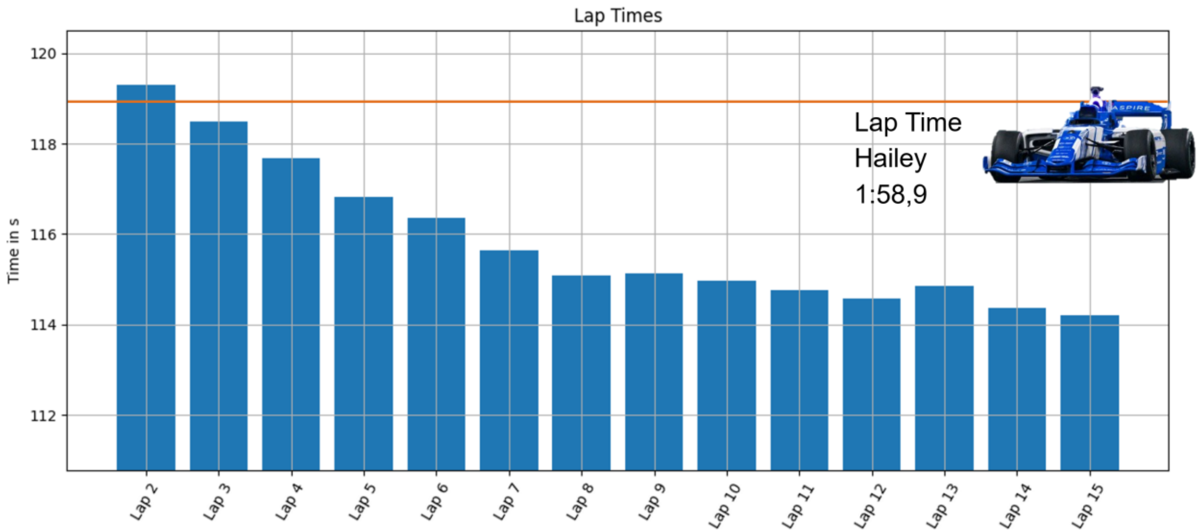


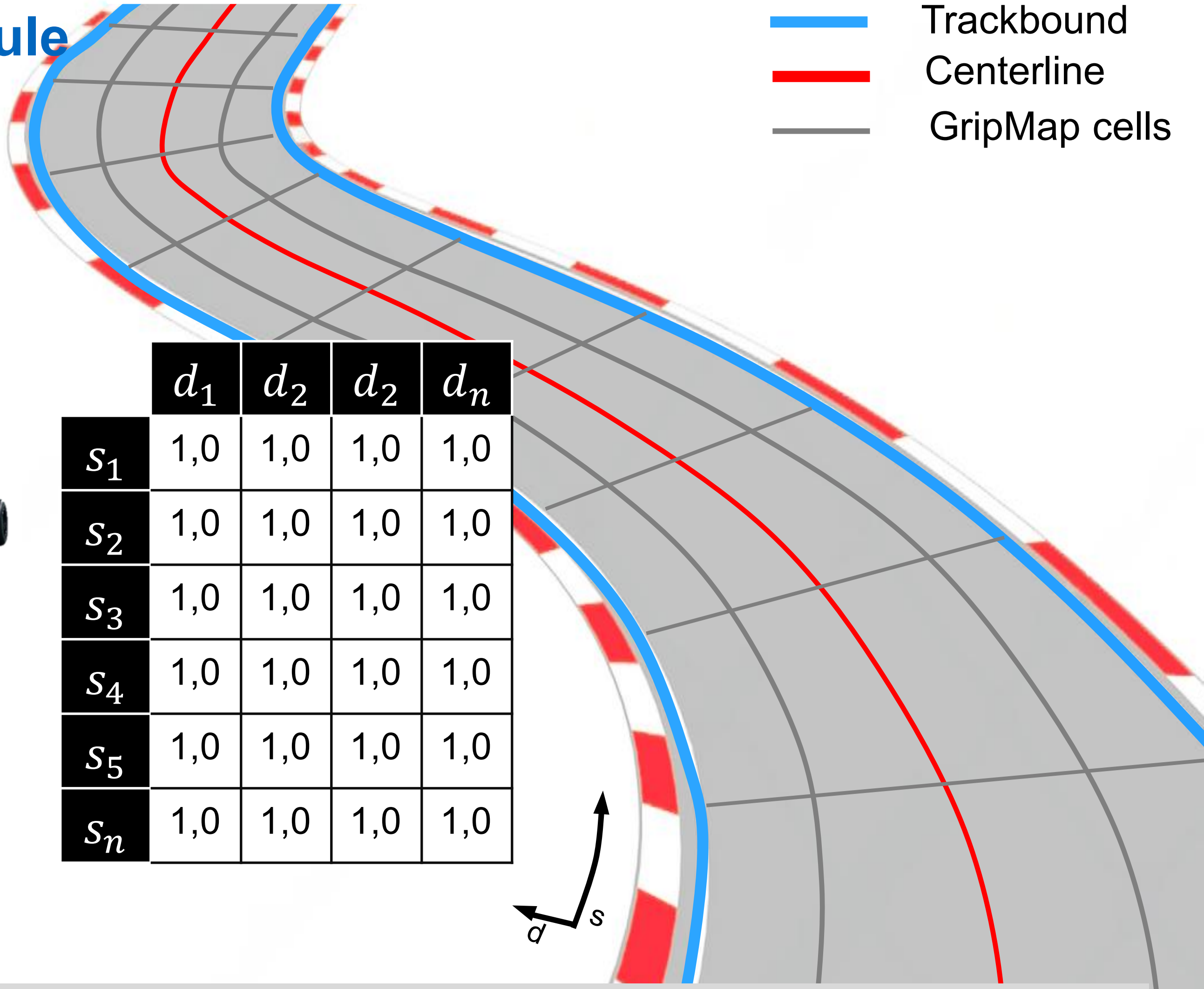


- Why? Derive a **feasible and dynamic trajectory**
- Design of a **sampling-based trajectory planner**
- Split into **offline and online** part
- **Dynamic overtaking,** real-time capable

Detecting the Vehicle Limits

Online Learning in Planning: GripMap Module





	d_1	d_2	d_2	d_n
s_1	1,0	1,0	1,0	1,0
s_2	1,0	1,0	1,0	1,0
s_3	1,0	1,0	1,0	1,0
s_4	1,0	1,0	1,0	1,0
s_5	1,0	1,0	1,0	1,0
s_n	1,0	1,0	1,0	1,0

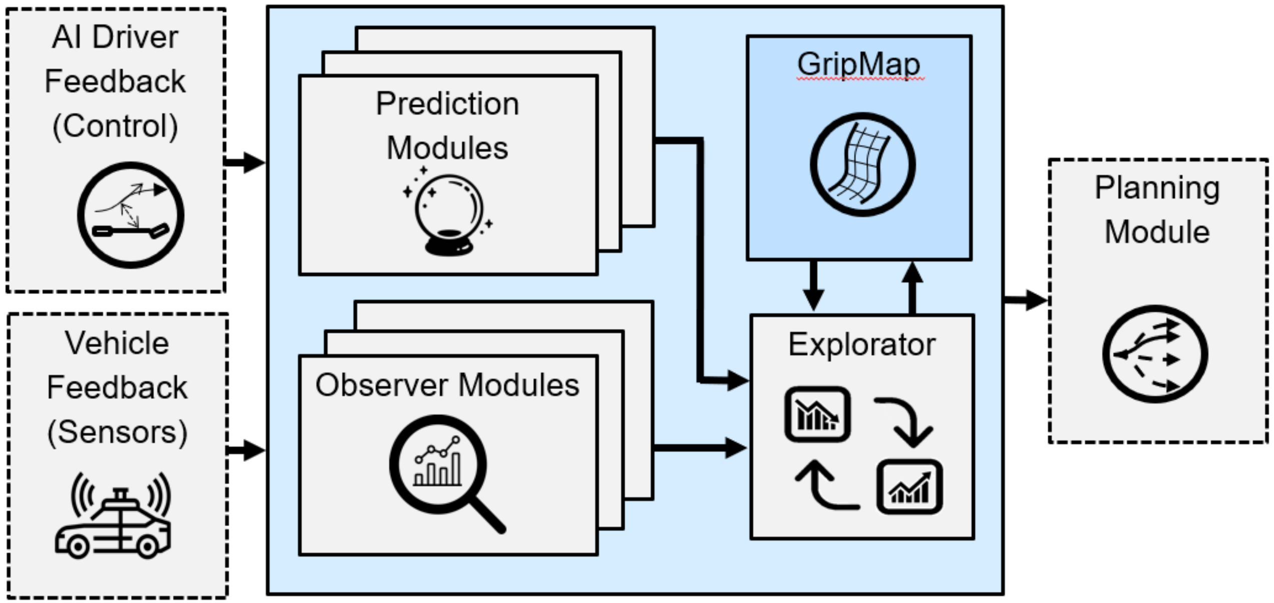
- Why? **Predict and estimate available vehicle acceleration potential**
- Create a map **for local differences** of grip scaling
- Use to plan an **adaptable, safe and reliable** trajectory

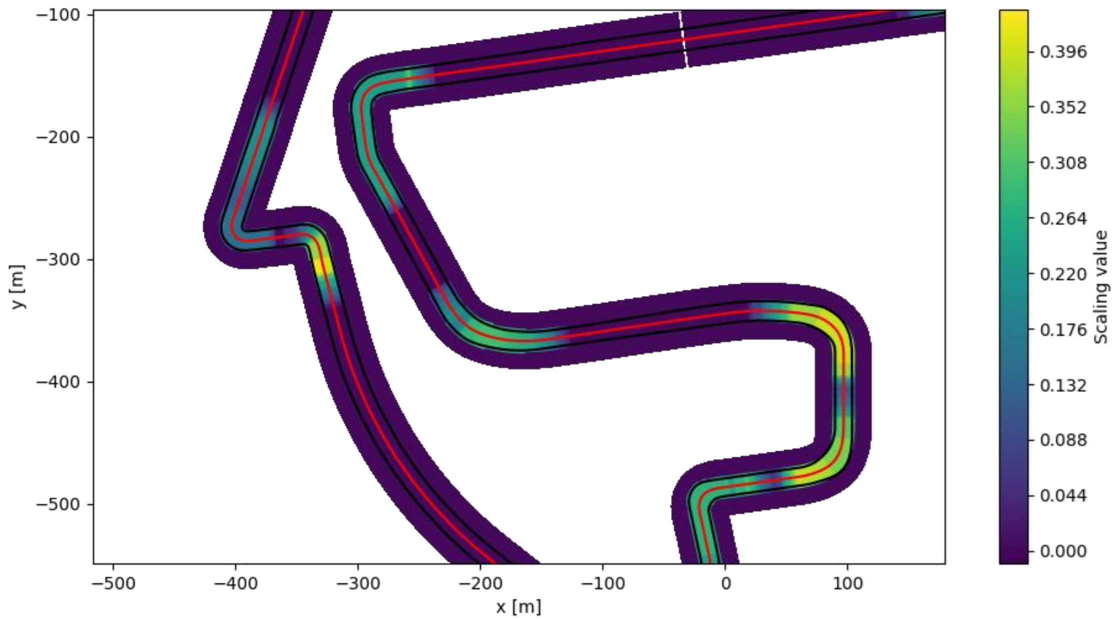
L. Hermansdorfer, J. Betz, M. Lienkamp, "A Concept for Estimation and Prediction of the Tire-Road Friction Potential for an Autonomous Racecar" in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019

F. Werner, R. Oberhuber, J.Betz: Accelerating "Autonomy: Insights from Pro Racers in the Era of Autonomous Racing -An Expert Interview Study", 2024 IEEE Intelligent Vehicles Symposium (IV)

Detecting the Vehicle Limits

Online Learning in Planning: GripMap Module





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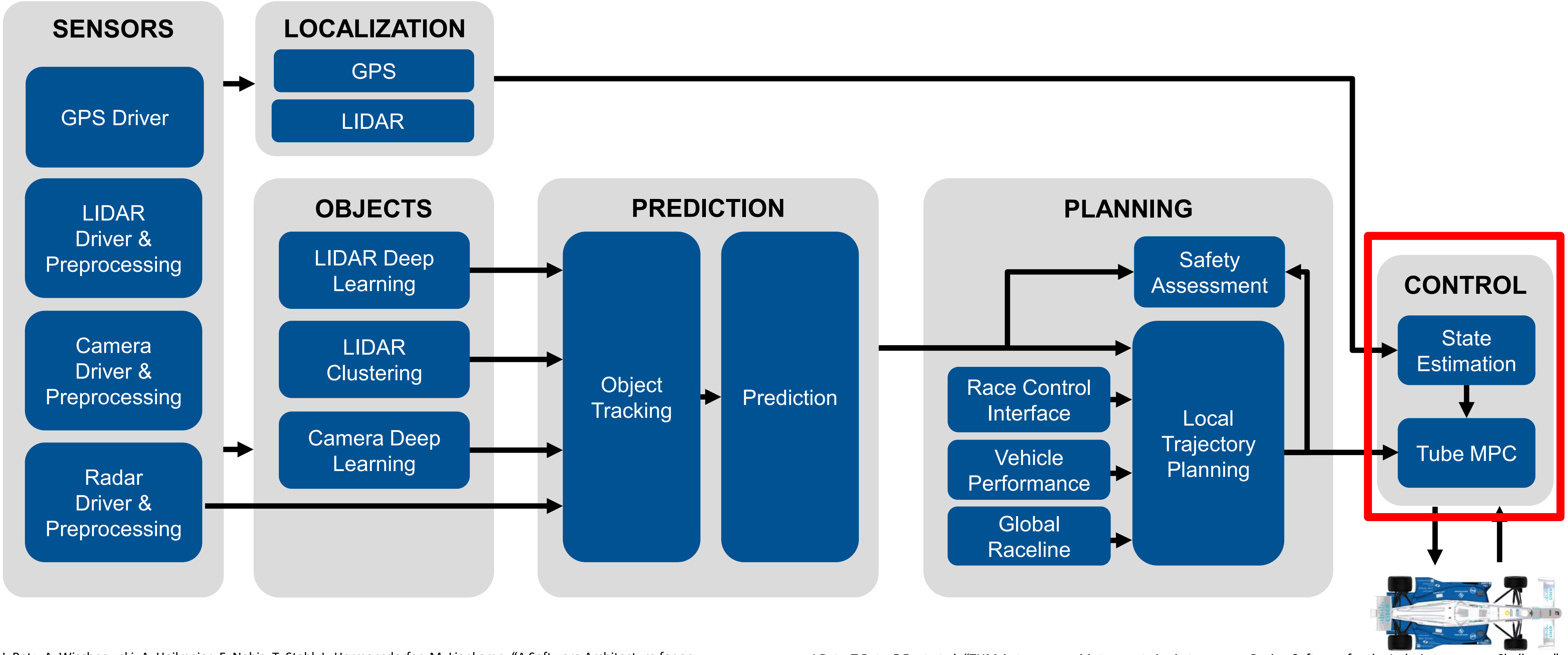
The Software Architecture

Autonomous Racing



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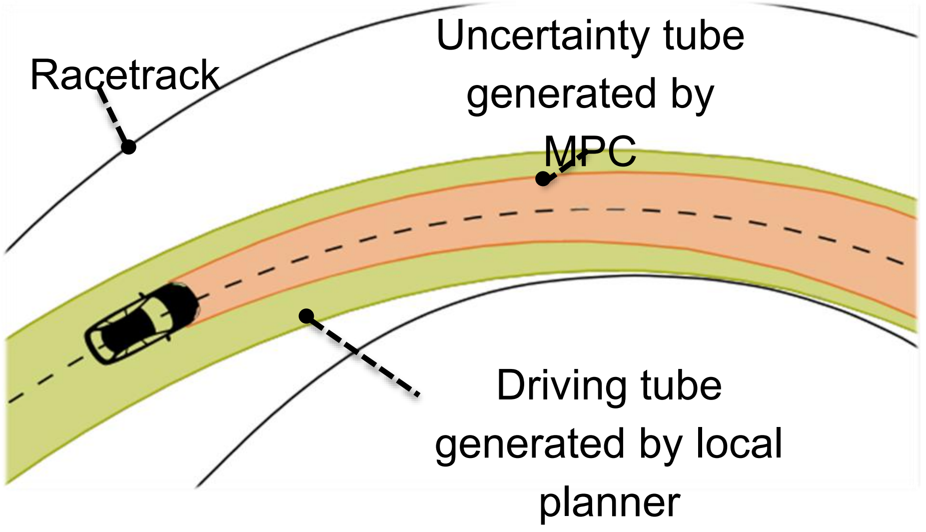
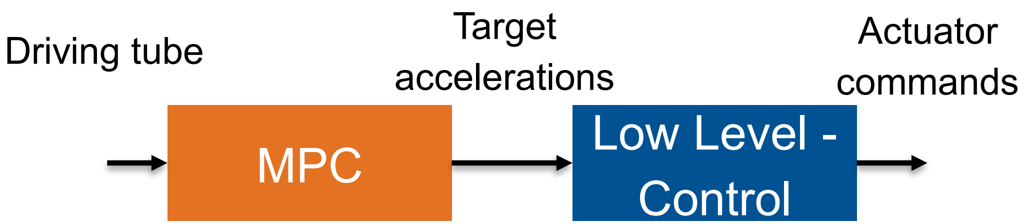
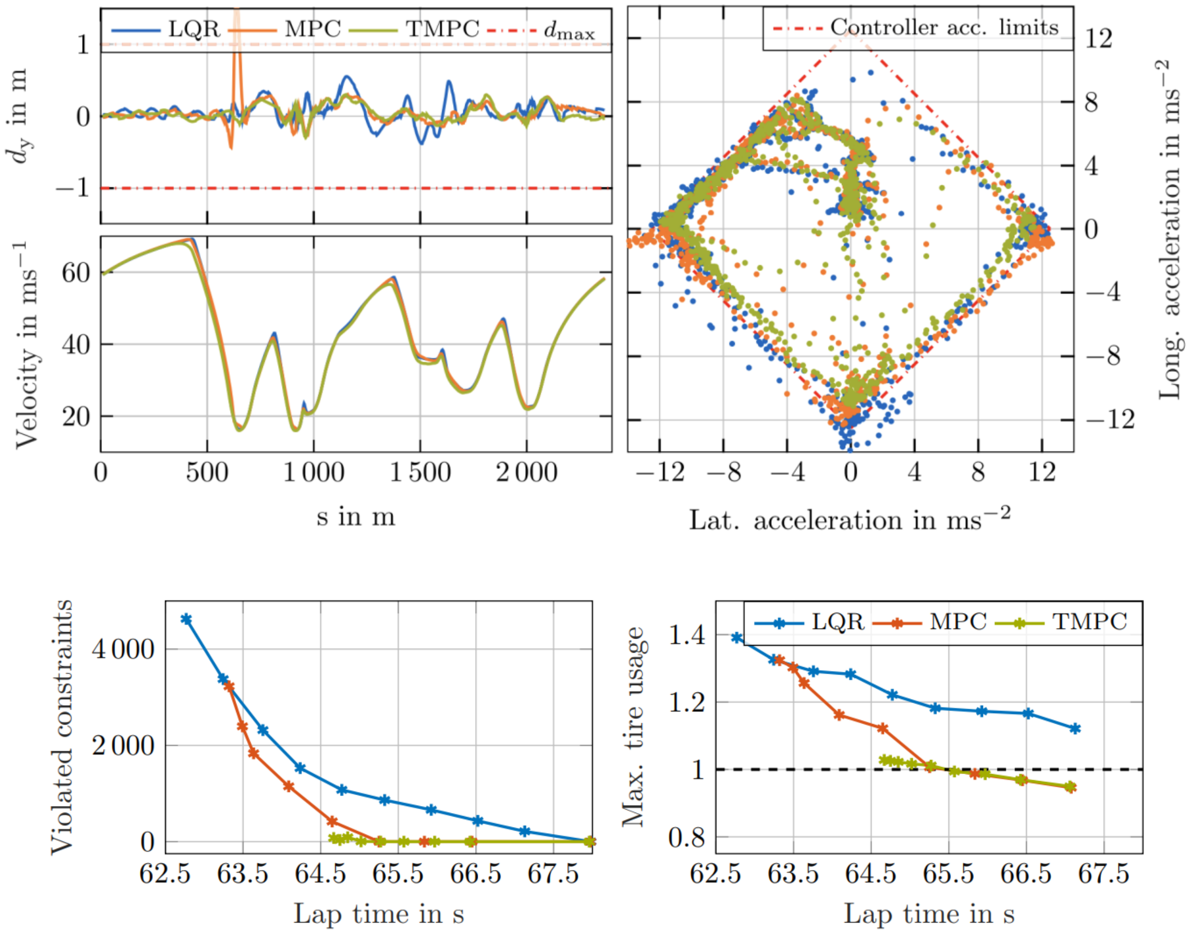
Motion Planning in Complex Scenarios
Control – Robust Tube MPC







- **Why?:** No constraint violation + fast
- **Simplifiy vehicle:** point-mass model + actuation dynamics
- **Combined** lateral / longitudinal tire limits
- **Smoothes trajectory** from planning
- **Inherit Robust design** improved via low-level control
- Capable of **running at 100Hz**

A. Wischnewski, M. Euler, S. Gümüs, and B. Lohmann, "Tube model predictive control for an autonomous race car," Vehicle System Dynamics, vol. 60, no. 9. Informa UK Limited, pp. 3151–3173, Jun. 23, 2021. doi: 10.1080/00423114.2021.1943461.

A. Wischnewski, T. Herrmann, F. Werner, and B. Lohmann, "A Tube-MPC Approach to Autonomous Multi-Vehicle Racing on High-Speed Ovals," IEEE Transactions on Intelligent Vehicles. Institute of Electrical and Electronics Engineers (IEEE), pp. 1–1, 2022. doi: 10.1109/tiv.2022.3169986.

- Autonomous Racing – The Future of Racing?
- How can we develop software for an Autonomous Racecar?
- **How do we improve software for an Autonomous Racecar?**
- Open Challenges & Future Efforts

Operate Everywhere, Every Time under Every Condition
Testing without a Vehicle

Early integration and full-stack testing

Continous integration via automated software tests and deployment

Semi-automated tuning of algorithm parameters

Continously improved simulations to challenge algorithms from the very beginning

Operate Everywhere, Every Time under Every Condition

Simulation

Operate Everywhere, Every Time under Every Condition

Bridging the Gap to the real World

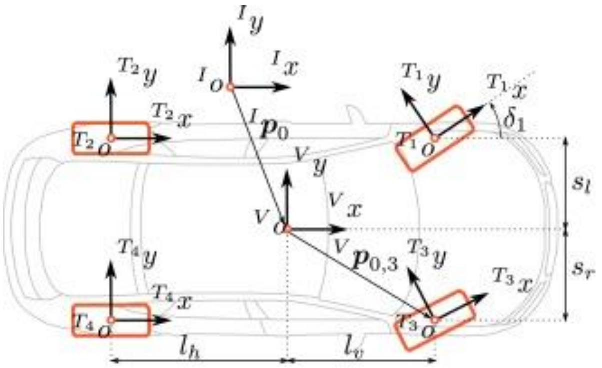
Autonomous Racing –
The Future of Racing?

How can we develop software for an
Autonomous Racecar?

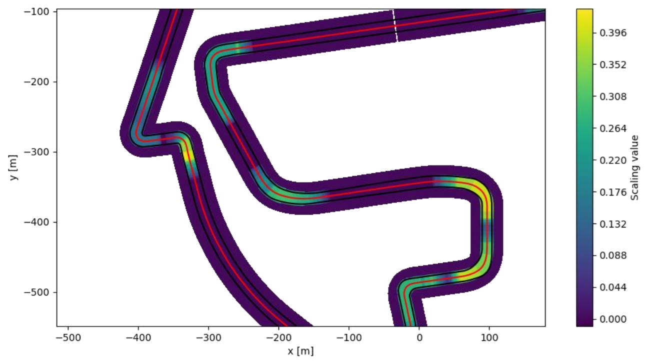
How do we improve software for an
Autonomous Racecar?

**Open Challenges &
Future Efforts**

Autonomous Racing
Open Challenges

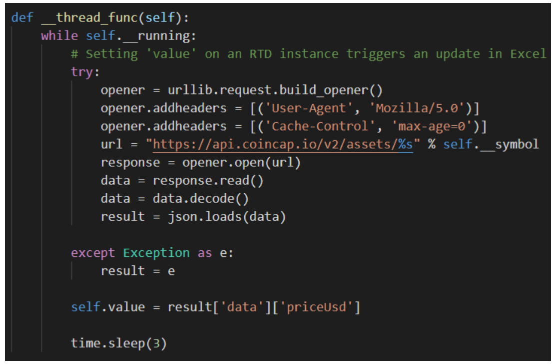


Vehicle Dynamic Knowledge



Self-Adapting Behavior

Control at and beyond the limits



Real-Time Capable Code

Sim to Real Gap

Working with real Hardware

J. Betz, H. Zheng, A. Liniger, U. Rosolia, P. Karle, M. Behl, V. Krovi, R. Mangharam, "Autonomous Vehicles on the Edge: A Survey on Autonomous Vehicle Racing", in *Open Journal of Intelligent Transportation Systems (OJ-ITS)*, 2022

Autonomous Racing
Open Challenges

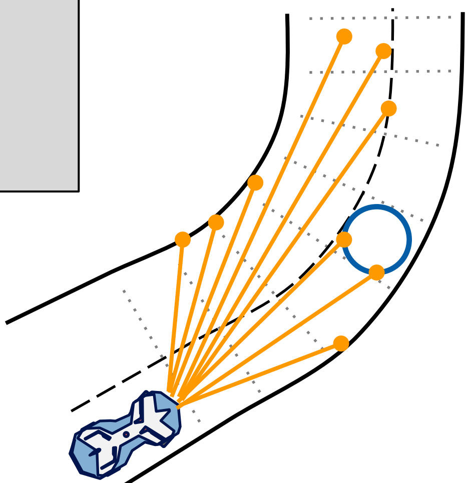
Perception

Localization

- (d) GPS
- Kalman Filter
- SLAM

Detection

- Fast R-CNN
- Point R-CNN
- SSD
- Yolo v4



Planning

Prediction

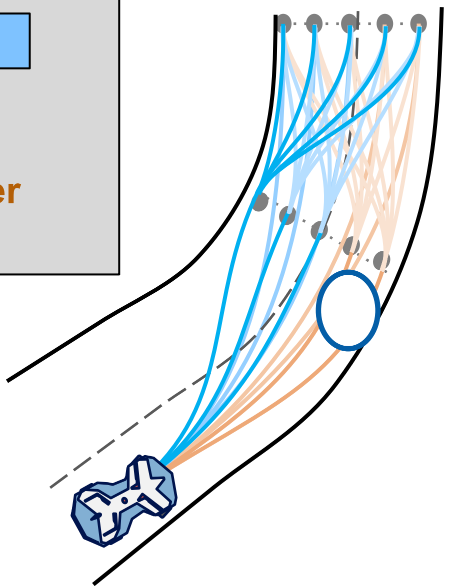
- Kalman Filter
- Deep Learning

Behavior

- State Machine
- Game Theory

Trajectory

- Optimization
- Frenet Planner

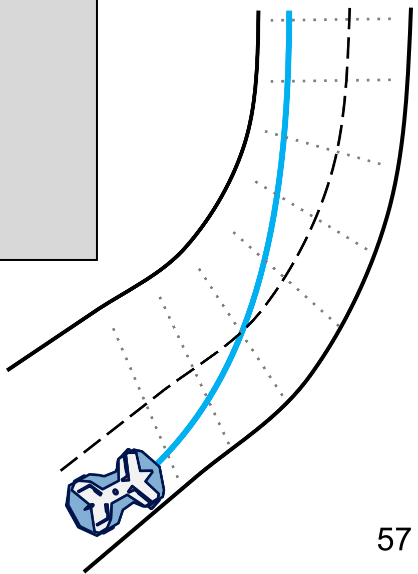


Control

Path

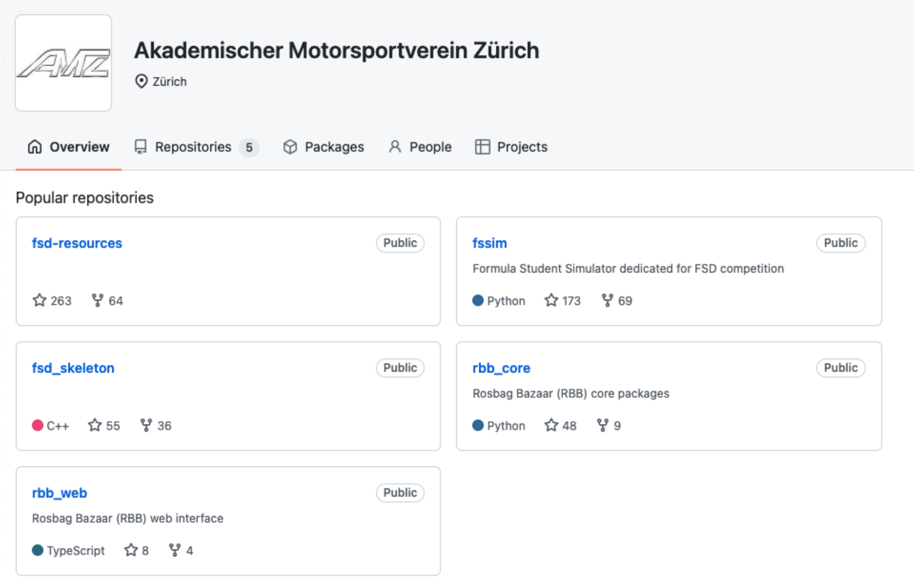
Velocity

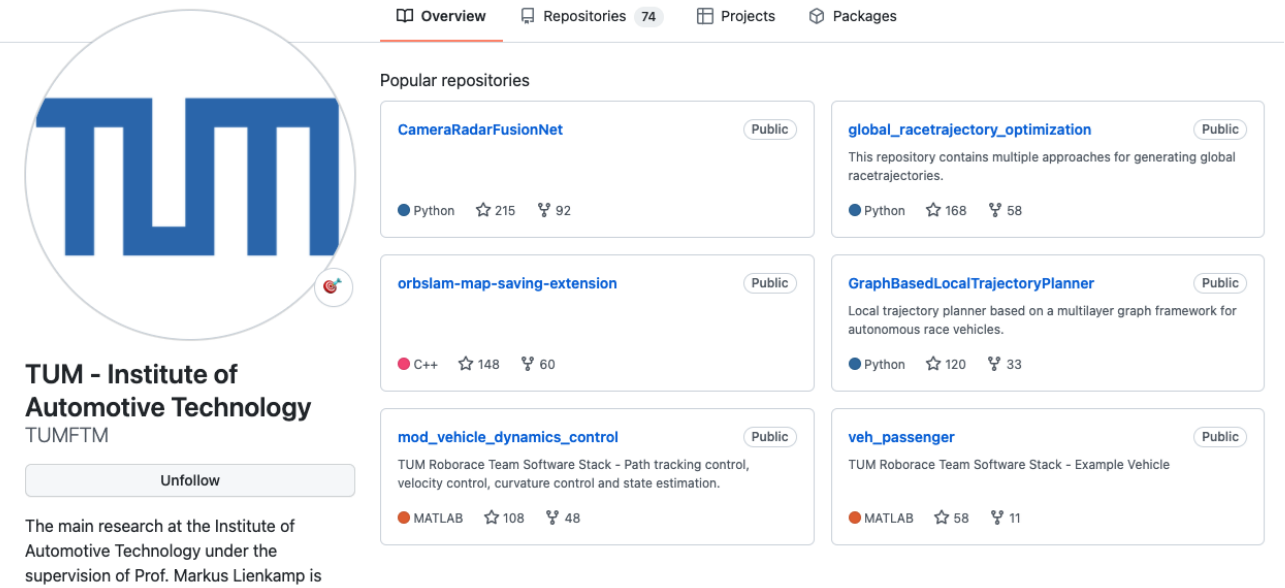
- PiD
- LQR
- MPC

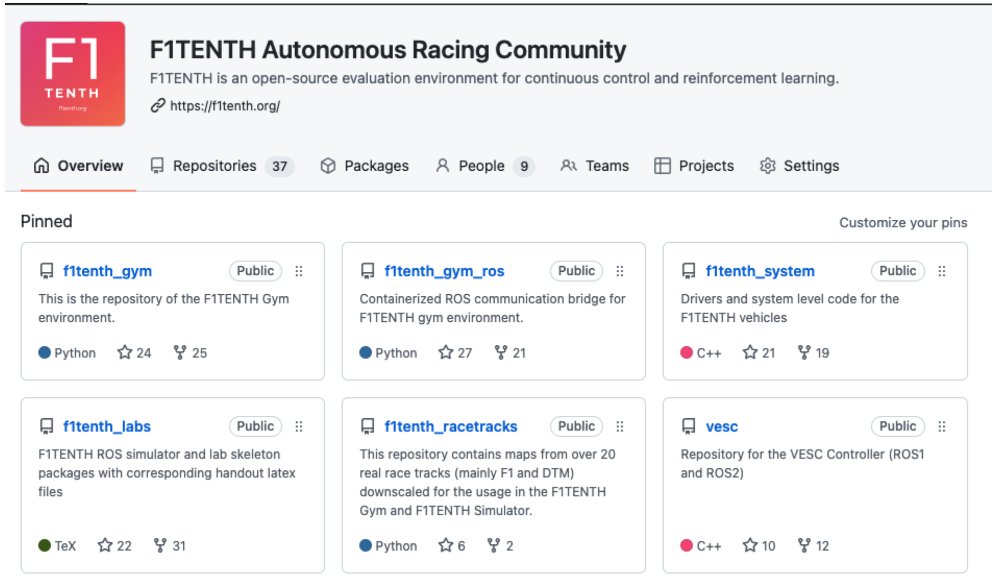


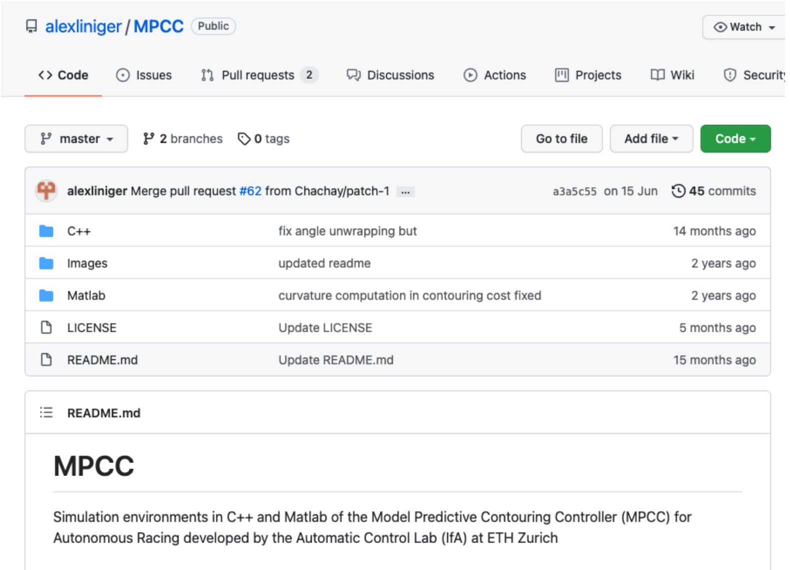
Autonomous Racing

Future Efforts – Open Source Software









Autonomous Racing
How to get involved

F1Tenth Hands-on

Startseite > Lehre

F1TENTH: Autonomous Driving Hands-on

Vortragende/r (Mitwirkende/r)	Johannes Betz [L], Felix Jahncke
Nummer	0000001440
Art	Praktikum
Umfang	4 SWS

Autonomous Racing
How to get involved

TUM Autonomous Motorsport

Topics:

- Perception, Prediction, Planning, Control, Localization, Grip Estimation
- Regular Thesis
- Extended Core Team Thesis

Thank you for your attention!

Join the Team!